

Targeting Persistent Disagreement: Coarse Interventions in Social Networks*

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Abstract

Institutions that shape public opinion can often change intervention intensity more easily than whom they reach. We study a decision-maker who commits to one exposure rule and reuses it. In a static problem this restriction does not bind. Dynamically it matters: intervention changes the opinions on which the same rule will later act. Optimal exposure follows the disagreement the network preserves and the agents with leverage over future opinions. An arbitrarily small perturbation can raise the number of exposure directions needed to replicate unrestricted targeting from two – a consensus direction and one disagreement direction – to one per agent. Yet near a benchmark where disagreement directions persist at the same rate and leverage is weakly aligned with current disagreement, the loss from restricting targeting is only second order. Using Yale climate opinions and Facebook social connectedness, our baseline application finds that one reusable rule captures 99.6% of attainable gains even after removing average opinion, yet reuse substantially reallocates exposure across states.

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1 Introduction

Institutions that shape public opinion can often change intervention intensity more easily than whom they reach. Health agencies, political campaigns, and social-media platforms operate through exposure rules – which regions receive more advertising, which communities see more prominent messages, which content is amplified – that fix the relative pattern of exposure while overall intensity adjusts with budgets, enforcement effort, or ranking salience. Meta’s third-party fact-checking program illustrates the distinction: changing its partner organizations and language coverage required organizational and policy changes, while Meta could adjust the ranking and labeling treatment applied to flagged content more readily (Facebook, 2016; Meta, 2025a,b; Gorwa, 2019; Krishnan et al., 2021).

We study this asymmetry between exposure pattern and intensity as *coarse targeting* in an opinion network. The decision-maker minimizes the discounted sum of opinion deviations from her agenda and intervention costs. She commits at the outset to an exposure vector $\mathbf{b} \in \mathbb{R}^n$ and then chooses a scalar intensity r_t in each period. This is a problem of instrument simplicity: one exposure pattern replaces a multidimensional intervention that could otherwise be redesigned each period. In a static problem, the restriction does not bind – any vector intervention $\mathbf{u}_0$ can be written as $\mathbf{b}r_0$. Its entire content is intertemporal: the same rule must be reused.

Reuse makes the problem endogenous. The rule shapes the opinion path on which it will later act, so it must be chosen against the path it creates. Two questions follow. What should a reusable rule target? And how much is lost by fixing the rule rather than retargeting each period? Together, these questions speak to a broader problem: when can a high-dimensional targeting problem be handled well with a simple instrument?

The first question is not answered by the most extreme opinions today or simply by how much weight others place on each agent. Consider four agents divided into two symmetric islands, $\{1, 2\}$ and $\{3, 4\}$. Each agent places weight 0.4 on each opinion in her own island and 0.1 on each opinion in the other island, so all four agents have equal centrality. Suppose the initial opinions are

$$\mathbf{x}_0 = (100, -99, 1, 1)'.$$

Agents 1 and 2 differ by 199, while the average opinions of the two islands differ by only 0.5. Social updating erases the within-island gap after one update, and optimal exposure

therefore assigns agents 1 and 2 identical exposure, regardless of how far apart their opinions are. What survives is the much smaller disagreement across islands: the extreme opinions in island 1 largely offset each other, while the common opinion in island 2 remains part of the path the network carries forward. Optimal targeting tilts toward island 2. A reusable rule targets disagreement by how much of it the network preserves, not by its current magnitude.

The example isolates the first force: persistence. With equal centrality – each agent receives the same aggregate weight from others – uniform exposure is optimal exactly when updating eliminates inherited disagreement, even in an asymmetric network. If disagreement survives, exposure follows it; near a symmetric benchmark, more persistent disagreement attracts more exposure. Reuse tempers this response: a reusable rule tilts less toward persistent disagreement than a one-period rule because the same exposure pattern remains in place after that disagreement has faded.

When centrality differs, a second force appears: leverage. Near the same benchmark, differences in leverage propagate along network paths in Katz–Bonacich form, so optimal exposure responds to how leverage propagates rather than to raw centrality. The opinion-leader network makes both implications sharp: greater leverage attracts more exposure, but the optimal rule remains dispersed because concentrating on the leader creates temporary disagreement. Yet one fixed exposure direction already attains the value of full targeting, under which the decision-maker can redesign the intervention vector $\mathbf{u}_t$ each period (Li and Tan, 2026). In this network, unequal leverage changes whom to target without creating value from retargeting. Persistence determines what remains worth moving; leverage determines who can move it.

The value of targeting flexibility depends on how much the fixed rule differs from unrestricted targeting along the opinion path it creates. We make this exact: the value gap is the trace of the product of a path matrix, recording the discounted opinion states along that path, and an intervention-error matrix, recording where fixed and unrestricted targeting differ. A fixed rule can therefore attain the full-targeting value even when the policies differ off the realized path; equality requires only that they prescribe the same intervention at every date along the path induced by the fixed rule. Flexibility has little value when the realized path is concentrated in directions on which the policies agree.

This shifts the question to network structure: when does targeting remain low-dimensional? At a symmetric benchmark where all forms of disagreement persist at the same rate, the

full-targeting intervention path lies in only two directions – the consensus direction and the direction of inherited disagreement. A network may therefore have many agents yet require only a low-dimensional intervention space.

This exact dimensional simplicity is fragile. An arbitrarily small perturbation can raise the number of exposure directions needed to replicate full targeting from two to one per agent. Economic value is more robust: near a benchmark with common disagreement persistence and weak leverage–disagreement alignment, the loss from restricting targeting is only second order. Exact dimension can therefore explode even when the value of additional flexibility remains small. The same two forces govern this loss: how differently the network preserves disagreement, and whether leverage lines up with the disagreement present.

We illustrate these forces using U.S. state-level climate opinions and the Facebook Social Connectedness Index. We use Facebook social connectedness to construct the propagation matrix and climate opinions as the initial state. California has about 21% more authority than Texas, but Texas has the higher future-persistence score, which summarizes how disagreement propagates into a state’s future opinion. Authority and future persistence point in opposite directions, and California receives only about 1% more exposure under the reported reusable rule. Ohio provides the complementary comparison, with low authority but unusually high future persistence and more exposure than authority alone would suggest.

A simple reusable rule also performs almost as well as unrestricted targeting. Even after removing average opinion, the reusable rule captures 99.6% of attainable gains. This near-equivalence survives reshuffling disagreement across states, but weakens sharply in random networks with comparable persistence. Reuse nevertheless reallocates exposure substantially. Thus reuse can change whom to target even when a simple rule is nearly sufficient in value.

A simple instrument fails not because disagreement keeps arriving, but because it arrives in many new directions. Recurrent agent-level shocks inject disagreement in unpredictable directions and can eventually make retargeting valuable. A persistent bias repeatedly injects the same source and, in the highly symmetric benchmark, adds at most one relevant disagreement direction. Recurrence alone does not defeat a reusable rule; diffuse new disagreement does.

Related literature

A. Network targeting and reusable instruments. Work on network targeting largely studies one-shot interventions, emphasizing network position, centrality, and key-player measures (Galeotti and Goyal, 2009; Ballester et al., 2006). Galeotti et al. (2020) decompose a one-shot intervention into the principal components of a static network game. Reuse changes the problem: the value of targeting a disagreement direction depends on how long it survives and on the opinion path created by the exposure rule itself.

In dynamic opinion models, interventions are typically tailored across agents or adjusted as opinions evolve (Ghaderi and Srikant, 2014; Candogan et al., 2022); Sadler (2023) studies how stochastic opinion fluctuations shape the concentration of influence campaigns in a network with stubborn agents. Closest in environment is Li and Tan (2026), which shares the linear-updating structure but studies how competing influencers with opposing agendas choose interventions in equilibrium. Here a single decision-maker commits to one exposure direction and reuses it, so the object of interest is the value and design of the restriction itself rather than strategic interaction.¹

B. Simple instruments and exposure design. Economic institutions often use low-dimensional instruments even when finer tailoring is feasible. Retail chains set nearly uniform prices across heterogeneous markets (DellaVigna and Gentzkow, 2019); common reserve prices are optimal in the canonical i.i.d. regular auction environment and remain approximately optimal when bidders differ (Myerson, 1981; Hartline and Roughgarden, 2009); and Carroll (2015) rationalizes linear contracts under robustness. Classical instrument-counting arguments ask how many instruments are needed for a given number of policy targets (Tinbergen, 1952). Our restriction is different: one exposure pattern is chosen initially and then reused. Exact replication may require many directions, but dimension counting is not the endpoint; we ask how much value those additional directions have along the endogenous path created by the reusable rule.

In digital environments, fine personalization is increasingly feasible but can also be constrained. Browsing data permit detailed retargeting (Lambrecht and Tucker, 2013), while privacy regulation can restrict the information available for targeting (Goldfarb and Tucker, 2011). We abstract from the source of coarseness and ask what a reusable exposure rule

¹Relatedly, Gaitonde et al. (2020) and Mostagir et al. (2022) examine how network structure shapes vulnerability to disagreement and misinformation.

should target and how much is lost by restricting intervention to such a rule.

C. Opinion dynamics. We use linear updating as a tractable representation of social interaction (DeGroot, 1974; Friedkin and Johnsen, 1990; DeMarzo et al., 2003; Golub and Jackson, 2010). Our focus is not on whether opinions converge, but on how an institution should intervene when current exposure changes the future value of influence. Richer models with behavioral or informational frictions can often be approximated locally by linear updating (Li and Tan, 2020; Molavi et al., 2018; Li and Tan, 2021), so the same propagation logic applies locally in those environments.

D. Dynamic systems and intervention. In engineering and applied mathematics, a large literature studies how intervention changes the evolution of network outcomes, often emphasizing feasibility, controllability, or stabilization (Lin, 1974; Pasqualetti et al., 2014; Liu and Barabási, 2016). Some recent work compares or selects where intervention enters the network (Summers et al., 2016; Lindmark and Altafini, 2018, 2020; Gao et al., 2021; Klickstein and Sorrentino, 2023). Our question instead concerns the design and value of a reusable exposure rule relative to unrestricted retargeting.

2 Model

A network is represented by a weighted directed graph $G = (N, A)$, where $N = \{1, \dots, n\}$ and $A = [a_{ij}]$ is row-stochastic, with $a_{ij} \geq 0$ and $\sum_j a_{ij} = 1$ for every i .² The entry a_{ij} is the weight agent i places on agent j 's opinion. A node may represent an individual, group, region, or other unit at which exposure is implemented. In Section 6, nodes are U.S. states.

Time is discrete. Let $\mathbf{x}_t \in \mathbb{R}^n$ denote opinions at date $t = 0, 1, 2, \dots$. After observing $\mathbf{x}_0$, a decision-maker commits at $t = 0$ to an exposure rule $\mathbf{b} \in \mathbb{R}^n$.³ In each period, she observes $\mathbf{x}_t$ and chooses a scalar intervention level $r_t \in \mathbb{R}$. Opinions evolve according to

$$\mathbf{x}_{t+1} = A\mathbf{x}_t + \mathbf{b}r_t.$$

The rule $\mathbf{b}$ fixes the relative exposure pattern across agents, while r_t scales its magnitude

²We use boldface lowercase letters for vectors and uppercase letters for matrices. All vectors are column vectors.

³We maintain that A and $\mathbf{x}_0$ are known when the exposure rule is chosen. Bloch and Shabayek (2023) study targeting under uncertainty about network position.

and sign. The total intervention is $\mathbf{u}_t = \mathbf{b}r_t$.⁴

Exposure is measured relative to a no-intervention baseline. For a positive intensity, positive coordinates of $\mathbf{b}$ raise exposure above baseline and negative coordinates lower it. Thus $\mathbf{b}$ is a relative exposure rule, not an allocation of a fixed positive budget.⁵

The decision-maker cannot redesign this cross-sectional rule over time. This *coarse targeting* restriction is entirely intertemporal. Any single-period intervention $\mathbf{u}_0$ can be written as $\mathbf{b}r_0$ with $\|\mathbf{b}\|=1$.

We impose $\|\mathbf{b}\| \leq 1$, a capacity bound. Increasing $\|\mathbf{b}\|$ permits the same intervention path to be implemented with smaller scalar intensities and hence at lower cost. Lemma 2 shows that the bound is attained whenever intervention is strictly valuable.⁶

Fix $\mathbf{b}$, a discount factor $\delta \in (0, 1)$, and a cost parameter $c > 0$. The decision-maker chooses $\{r_t\}_{t \geq 0}$ to solve

$$\max_{\{r_t\}_{t \geq 0}} - \sum_{t=0}^{\infty} \delta^t (\mathbf{x}'_t Q \mathbf{x}_t + cr_t^2),$$

where $\mathbf{x}'_t Q \mathbf{x}_t$ is the loss from deviations from the decision-maker's agenda and cr_t^2 is the cost of scalar intervention. We normalize the decision-maker's common agenda to $\mathbf{0}$. This is without loss because $A\mathbf{1} = \mathbf{1}$: any common target can be removed by recentering opinions. The matrix Q is symmetric and positive definite and weights opinion deviations. We allow general Q unless otherwise stated; Section 4 sets $Q = I$, while Section 6 uses population weights.

As a benchmark, under *full targeting* the decision-maker chooses an unrestricted intervention vector $\mathbf{u}_t \in \mathbb{R}^n$ in each period:

$$\mathbf{x}_{t+1} = A\mathbf{x}_t + \mathbf{u}_t,$$

with intervention cost $c\|\mathbf{u}_t\|^2$. When $\|\mathbf{b}\|=1$, a coarse intervention $\mathbf{u}_t = \mathbf{b}r_t$ has the same cost, $c\|\mathbf{u}_t\|^2 = cr_t^2$. The two technologies therefore differ only in the ability to redesign the exposure pattern over time. Let $v^f(\mathbf{x}_0)$ denote the full-targeting value. Section 5

⁴The linear updating rule can also be interpreted locally. For a smooth rule $\mathbf{x}_{t+1} = F(\mathbf{x}_t) + \mathbf{b}r_t$, deviations around a baseline $\bar{\mathbf{x}}$ satisfy $\Delta\mathbf{x}_{t+1} \approx J(\bar{\mathbf{x}})\Delta\mathbf{x}_t + \mathbf{b}r_t$. The relevant local propagation matrix is therefore $J(\bar{\mathbf{x}})$; see Appendix S3.

⁵Because $(\mathbf{b}, r_t)$ and $(-\mathbf{b}, -r_t)$ generate the same intervention, the rule is identified only up to sign.

⁶The Euclidean bound is invariant to orthonormal changes of basis. Other institutional restrictions, such as nonnegativity, may generate corner solutions.

characterizes the value of this targeting flexibility.

For a fixed exposure rule $\mathbf{b}$, let $v(\mathbf{x}_0; \mathbf{b})$ denote the value obtained when intervention levels are chosen optimally. The decision-maker chooses $\mathbf{b}$ to solve

$$\max_{\|\mathbf{b}\| \leq 1} v(\mathbf{x}_0; \mathbf{b}). \quad (1)$$

The exposure rule must therefore be chosen against the opinion path that it itself induces.

The coarse restriction is meant to capture settings in which intervention intensity can adjust readily, while the cross-sectional pattern of exposure is harder to redesign. Two examples are useful to keep in mind.

Platform ranking. Nodes are communities or content categories, and $\mathbf{b}$ fixes their relative amplification or down-ranking from baseline. Changing the pattern requires product, policy, or compliance review, while its overall salience r_t adjusts more frequently.

Regulatory attention. Nodes are jurisdictions or firms, and $\mathbf{b}$ specifies how enforcement or outreach is reallocated from baseline. The pattern changes slowly, while the overall intensity of inspections, audits, or outreach responds to current conditions.

3 Characterizing optimal coarse targeting

We begin with the one-period benchmark. Suppose opinions evolve once, $\mathbf{x}_1 = A\mathbf{x}_0 + \mathbf{b}r_0$, with $\|\mathbf{b}\| \leq 1$. For any fixed exposure rule $\mathbf{b}$, the optimal scalar intervention r_0 is linear in $\mathbf{x}_0$. Since only the total intervention $\mathbf{u}_0 = \mathbf{b}r_0$ matters, and any one-period intervention can be implemented with $\|\mathbf{b}\| = 1$, the static problem is equivalent to choosing $\mathbf{u}_0$ directly. If $QA\mathbf{x}_0 \neq \mathbf{0}$, the optimum is

$$\mathbf{u}_0^* = \mathbf{b}^* r_0 = -\delta(cI + \delta Q)^{-1} QA\mathbf{x}_0.$$

Thus fixing an exposure direction is costless in one period.

The restriction becomes substantive only when the same direction must be reused. For a fixed $\mathbf{b} \neq \mathbf{0}$, the intervention problem has a quadratic value,

$$v(\mathbf{x}; \mathbf{b}) = -\mathbf{x}' K^*(\mathbf{b}) \mathbf{x}.$$

The Bellman equation is

$$v(\mathbf{x}; \mathbf{b}) = \max_r \left\{ -\mathbf{x}'Q\mathbf{x} - cr^2 - \delta(A\mathbf{x} + \mathbf{b}r)'K^*(\mathbf{b})(A\mathbf{x} + \mathbf{b}r) \right\}. \quad (2)$$

The optimal intervention level is linear,

$$r^*(\mathbf{x}; \mathbf{b}) = -\delta(c + \delta\mathbf{b}'K^*(\mathbf{b})\mathbf{b})^{-1}\mathbf{b}'K^*(\mathbf{b})A\mathbf{x} \equiv -\mathbf{p}(\mathbf{b})'\mathbf{x}, \quad (3)$$

and substitution gives the fixed-point equation

$$K^*(\mathbf{b}) = Q + \delta A' \left(K^*(\mathbf{b}) - \delta K^*(\mathbf{b})\mathbf{b}(c + \delta\mathbf{b}'K^*(\mathbf{b})\mathbf{b})^{-1}\mathbf{b}'K^*(\mathbf{b}) \right) A. \quad (4)$$

We first use a standard discounted LQ result to fix the continuation-value and stability objects used below.

Lemma 1. *For every fixed $\mathbf{b} \neq \mathbf{0}$, $c > 0$, and $\delta \in (0, 1)$, there exists a unique symmetric positive definite matrix $K^*(\mathbf{b})$ solving (4). The optimal intervention level is given by (3), and the induced transition matrix $\tilde{A}(\mathbf{b})$ satisfies $\rho(\sqrt{\delta} \tilde{A}(\mathbf{b})) < 1$.*

The induced transition matrix $\tilde{A}(\mathbf{b}) \equiv A - \mathbf{b}\mathbf{p}(\mathbf{b})'$ governs opinion dynamics, with $\mathbf{x}_{t+1} = \tilde{A}(\mathbf{b})\mathbf{x}_t$. The exposure rule thus determines how the decision-maker intervenes, and also shapes the opinion path on which the same rule will later be used.

Lemma 2. *Suppose $QAx_0 \neq \mathbf{0}$. Then every optimal exposure rule satisfies $\|\mathbf{b}^*\| = 1$.*

Scaling up a fixed exposure direction implements the same intervention path with lower scalar-intervention cost. Since $QAx_0 \neq \mathbf{0}$ makes intervention strictly valuable, every optimum exhausts the capacity bound. The outer problem is therefore

$$\max_{\|\mathbf{b}\|=1} v(\mathbf{x}_0; \mathbf{b}) = -\mathbf{x}_0'K^*(\mathbf{b})\mathbf{x}_0, \quad K^*(\mathbf{b}) \text{ solves (4)}. \quad (5)$$

The object linking the two roles of the exposure rule is the discounted opinion-path matrix

$$Z(\mathbf{b}) = \sum_{t=0}^{\infty} (\sqrt{\delta} \tilde{A}(\mathbf{b}))^t \mathbf{x}_0 \mathbf{x}_0' (\sqrt{\delta} \tilde{A}(\mathbf{b}))^t = \sum_{t=0}^{\infty} \delta^t \mathbf{x}_t \mathbf{x}_t'. \quad (6)$$

It records where the path induced by $\mathbf{b}$ places its discounted weight: directions that are large or persistent receive greater weight. Standard perturbation results imply that $K^*(\mathbf{b})$,

and hence $v(\mathbf{x}_0; \mathbf{b})$, vary smoothly with $\mathbf{b}$ and the primitives over the maintained parameter region (see Konstantinov et al. (1993) among others).

Proposition 1. *An optimal exposure rule exists. If $QA\mathbf{x}_0 \neq \mathbf{0}$, any optimal exposure rule $\mathbf{b}^*$ satisfies, for some multiplier μ on $\|\mathbf{b}\| = 1$,*

$$\mu \mathbf{b}^* = K^*(\mathbf{b}^*) \tilde{A}(\mathbf{b}^*) Z(\mathbf{b}^*) \mathbf{p}(\mathbf{b}^*). \quad (7)$$

Proposition 1 captures the fixed point at the heart of coarse targeting.⁷ The path matrix $Z(\mathbf{b}^*)$ weights the directions that matter, while the remaining terms in (7) translate them into the value of reaching particular agents. The same path will later determine the value of additional targeting flexibility.

The optimal exposure direction is homogeneous in initial opinions: rescaling $\mathbf{x}_0$ by any nonzero scalar changes intervention intensity but not the optimal direction.

Because changing the exposure rule changes the path that gives the rule its value, the outer problem need not be concave. Nevertheless, under the nondegeneracy condition in Corollary S1, its stationary points are isolated and finite, and every local maximum is strict. Thus nonconcavity generates a finite collection of isolated local candidates rather than continua of optima, making multi-start search a natural numerical approach.

4 Separating persistence from leverage

A reusable rule should respond to disagreement for two reasons: disagreement may persist, or it may shift average opinion because some agents carry more influence than others. We call these forces persistence and leverage. Throughout this section, let $Q = I$.

Let $\mathbf{u} = \mathbf{1}/\sqrt{n}$ be the uniform direction, $\Pi = I - \mathbf{u}\mathbf{u}'$ the projection onto disagreement, and $\mathbf{d} = \Pi\mathbf{x}_0$ the initial disagreement. Define

$$A_{\perp} = \Pi A \Pi,$$

⁷The characterization extends to substochastic matrices A satisfying $\rho(A) < 1$. Row stochasticity is used in Section 4 to separate average opinion from disagreement.

the part of updating that maps disagreement back into disagreement. Leverage is

$$\ell = A'\mathbf{u} - \mathbf{u},$$

so ℓ_i measures the excess aggregate weight placed on agent i relative to the uniform benchmark. Since $\mathbf{d} \perp \mathbf{u}$,

$$A\mathbf{d} = \mathbf{u}(\ell'\mathbf{d}) + A_\perp\mathbf{d}.$$

The first term shifts average opinion; the second is surviving disagreement. Absent intervention, $\Pi\mathbf{x}_t = A_\perp^t\mathbf{d}$, so $A_\perp$ governs the persistence of disagreement.

4.1 Persistent disagreement under equal centrality

In a row-stochastic network, agents are equally central when $\mathbf{1}'A = \mathbf{1}'$, or equivalently, when A is doubly stochastic (each column also sums up to one). In the decomposition above, this is exactly $\ell = \mathbf{0}$: disagreement cannot shift average opinion, so only its persistence under $A_\perp$ matters. This condition allows directed and asymmetric interactions. Let the average initial opinion be $\bar{x}_0 = \mathbf{1}'\mathbf{x}_0/n$.

Proposition 2. *Suppose A is doubly stochastic and $Q = I$, and suppose $\bar{x}_0 \neq 0$. Then uniform targeting is optimal if and only if $A\mathbf{d} = \mathbf{0}$.*

Under equal centrality, persistent disagreement is the only source of nonuniform targeting. Uniform exposure moves average opinion without creating disagreement.⁸ If $A\mathbf{d} = \mathbf{0}$, inherited disagreement disappears immediately; if $A\mathbf{d} \neq \mathbf{0}$, a nonuniform rule can move disagreement that survives.

The remainder of this subsection focuses on symmetric networks. Let

$$A = PDP', \quad D = \text{diag}(\lambda_1, \dots, \lambda_n), \quad \mathbf{p}_1 = \mathbf{u},$$

where we take the uniform direction as the first eigenvector, and let $\tilde{\mathbf{x}} = P'\mathbf{x}$ and $\tilde{\mathbf{b}} = P'\mathbf{b}$. The first coordinate is average opinion; each $j > 1$ is a disagreement dimension. The magnitude $|\lambda_j|$ measures how slowly disagreement in dimension j decays, while the sign of

⁸The assumption $\bar{x}_0 \neq 0$ ensures that there is an average-opinion component for uniform exposure to target.

λ_j determines whether its direction is preserved or reversed.⁹

Lemma 3. *Suppose A is symmetric and $Q = I$, and suppose $Ax_0 \neq 0$. In any orthonormal basis of eigenvectors of A , if $\lambda_j \tilde{x}_0^j = 0$ for some $j > 1$, then every optimal exposure rule satisfies $\tilde{b}_j^* = 0$.*

Lemma 3 rules out exposure to disagreement that is either absent initially or eliminated after one update. If several disagreement dimensions are carried forward in exactly the same way, their relative proportions never change. Applying the lemma in a basis aligned with the combination present in initial opinions then rules out exposure in the remaining directions.

The two-island example in the introduction is a direct application. Between-island disagreement survives social updating, while within-island disagreement disappears after one round. Lemma 3 therefore implies identical exposure within each island, regardless of the magnitude of within-island disagreement.

A highly symmetric family makes the implication transparent:

$$A(\rho) = \rho I + (1 - \rho) \frac{J}{n}, \quad \rho \in \left[-\frac{1}{n-1}, 1 \right]. \quad (8)$$

Here $A(\rho)\mathbf{d} = \rho\mathbf{d}$, so absent intervention inherited disagreement follows the path $\rho^t\mathbf{d}$. Every disagreement direction is carried forward in the same way, so we can take $\mathbf{d}/\|\mathbf{d}\|$ as one of the disagreement eigenvectors. Lemma 3 then rules out exposure in every remaining disagreement direction, and hence

$$\mathbf{b}^*(\rho) \in \text{span}\{\mathbf{u}, \mathbf{d}\}.$$

Writing $\mathbf{b}^*(\rho) = \alpha\mathbf{u} + \beta\mathbf{d}/\|\mathbf{d}\|$, the induced dynamics reduce to

$$m_{t+1} = m_t + \alpha r_t, \quad y_{t+1} = \rho y_t + \beta r_t,$$

where $m_t = \mathbf{u}'\mathbf{x}_t$ and y_t is disagreement along $\mathbf{d}$. The fixed rule therefore divides exposure between average opinion and the direction of inherited disagreement. At $\rho = 0$, Proposition 2 implies that uniform exposure is optimal whenever intervention is valuable. At $\rho = 1$, $A(\rho) = I$, so every direction is carried forward unchanged. Taking $\mathbf{x}_0/\|\mathbf{x}_0\|$ as the first

⁹The change of basis rotates the state and exposure vector, not the instrument: the decision-maker still chooses one scalar intervention level r_t each period.

basis vector, Lemma 3 rules out exposure in every other direction, so an optimal exposure direction is proportional to $\mathbf{x}_0$.

4.2 Leverage without persistent disagreement

Definition 1. A network is an **opinion-leader network** if $A = \mathbf{1}\mathbf{a}'$, where $\mathbf{a}'\mathbf{1} = 1$, the weights are distinct, and agents are labeled so that $a_1 > \dots > a_n > 0$.

Every agent listens to the same authority-weighted average $\mathbf{a}'\mathbf{x}$, so all disagreement disappears after one update. In the decomposition above,

$$A_{\perp} = 0, \quad \ell = \sqrt{n} \left(\mathbf{a} - \frac{\mathbf{1}}{n} \right).$$

Persistence is absent, and differences in authority coincide with differences in leverage.

Proposition 3. Fix $\delta \in (0, 1)$ and $c > 0$. Suppose $Q = I$ and $\mathbf{a}'\mathbf{x}_0 \neq 0$, and normalize the sign of $\mathbf{b}^*$ so that $\mathbf{a}'\mathbf{b}^* > 0$. The optimal exposure direction is independent of $\mathbf{x}_0$ and has the form

$$\mathbf{b}^* = \alpha \mathbf{u} + \gamma \ell, \quad \alpha > \gamma > 0.$$

Consequently,

$$b_1^* > \dots > b_n^* > 0, \quad b_i^* - b_j^* = \gamma(\ell_i - \ell_j).$$

Here no disagreement survives one round. Under equal centrality that would make uniform exposure optimal; leverage is what breaks uniformity. The independence from $\mathbf{x}_0$ is stronger than scale invariance: after one update the network reduces every initial condition to the scalar $\mathbf{a}'\mathbf{x}_0$, so the same exposure direction is optimal whenever this scalar is nonzero. Because $\ell_i - \ell_j = \sqrt{n}(a_i - a_j)$, exposure rises with authority, while every agent receives positive exposure.

The rank-one structure of A maps every state into a single aggregate. Consequently, under full targeting all interventions lie in one fixed one-dimensional direction; only their intensity changes with the state. A fixed exposure rule can therefore reproduce full targeting exactly and attain its value. Unequal leverage changes whom to target, but by itself creates no value from retargeting.

4.3 Persistence and leverage together

We now combine persistence and leverage around a common benchmark. Let $A_0 = PDP'$ be symmetric and row-stochastic, with $\mathbf{p}_1 = \mathbf{u}$. By scale invariance, normalize the benchmark initial opinion to $\mathbf{e}_1$ and perturb opinions and the network according to

$$\tilde{\mathbf{x}}_0 = \mathbf{e}_1 + \boldsymbol{\xi}, \quad A_\varepsilon = A_0 + \varepsilon B, \quad B\mathbf{1} = \mathbf{0},$$

restricting attention to sufficiently small $\boldsymbol{\xi}$ and $|\varepsilon|$ such that A_ε remains nonnegative.

Define $\mathbf{s} = B'\mathbf{u}$. Since A_0 is symmetric, $\ell(A_\varepsilon) = \varepsilon \mathbf{s}$. Thus $\mathbf{s}$ is the first-order change in leverage induced by the network perturbation. Moreover, $\mathbf{u}'\mathbf{s} = 0$, so leverage is redistributed across agents without changing its total.

At the benchmark, uniform exposure is optimal. Let $\tau \in (0, 1)$ denote the induced persistence of the average-opinion coordinate under the optimal intensity policy, so that the average opinion evolves as $\tilde{x}_{t+1}^1 = \tau \tilde{x}_t^1$.

Proposition 4. *Fix (δ, c) . Suppose $Q = I$ and $\boldsymbol{\xi} \perp \mathbf{e}_1$, and define*

$$\mathbf{g} = (I - \delta\tau A_0')^{-1}\mathbf{s} = \sum_{j>1} g_j \mathbf{p}_j.$$

There exist functions $M : [-1, 1] \rightarrow \mathbb{R}$ and $\chi : [-1, 1] \rightarrow \mathbb{R}_{++}$, with M strictly increasing, such that the unique strict local optimum near $\mathbf{e}_1$ satisfies

$$\tilde{\mathbf{b}}^*(\boldsymbol{\xi}, \varepsilon) = \mathbf{e}_1 + \sum_{j>1} [M(\lambda_j)\xi_j + \varepsilon\chi(\lambda_j)g_j] \mathbf{e}_j + o(\|\boldsymbol{\xi}\| + |\varepsilon|).$$

Moreover, $M(0) = 0$, $M(1) = 1$, and $0 < M(\lambda) < \lambda$ for all $\lambda \in (0, 1)$.

Setting $\varepsilon = 0$ isolates persistence. In the one-period problem, dimension j receives exposure coefficient $\lambda_j \xi_j$:

$$\tilde{\mathbf{b}}^{\text{static}}(\boldsymbol{\xi}) = \mathbf{e}_1 + \sum_{j>1} \lambda_j \xi_j \mathbf{e}_j + o(\|\boldsymbol{\xi}\|).$$

Under reuse, the corresponding coefficient is $M(\lambda_j)\xi_j$. A reusable rule therefore tilts less toward positively persistent disagreement: the same exposure pattern remains in place as disagreement decays, while average opinion persists. The wedge is strict for $\lambda \in (0, 1)$ and

disappears at the endpoints. At $\lambda = 0$ the disagreement is immediately irrelevant, while at $\lambda = 1$ it is as persistent as average opinion.¹⁰ For $\lambda < 0$, updating reverses the direction of disagreement from one period to the next, so exposure tilts against its current direction.

The two-island example gives the same comparison away from the local perturbation. With $\delta = 0.9$ and $c = 10$, its between-island disagreement has eigenvalue $\lambda = 0.6$, and the island-2 to island-1 exposure ratio falls from 1.50 under the one-period rule to 1.15 under reuse. Because this disagreement decays while the exposure pattern remains fixed, reuse tempers the tilt toward it.

Setting $\xi = 0$ isolates leverage. The optimal rule responds not to the direct change in leverage $\mathbf{s}$ alone, but to its propagation through the network:

$$\mathbf{g} = \mathbf{s} + \delta\tau A'_0 \mathbf{s} + \delta^2\tau^2 (A'_0)^2 \mathbf{s} + \dots$$

Thus $\mathbf{g}$ measures effective leverage after accounting for its future propagation, with $\delta\tau$ reflecting discounting and the decline of average opinion under intervention. Persistence and leverage add to first order; their interaction enters only at higher order.

Perturbing the clique toward an opinion-leader network connects Proposition 4 to Proposition 3. Let

$$A_\varepsilon = \mathbf{1}\mathbf{a}'_\varepsilon, \quad \mathbf{a}_\varepsilon = (1 - \varepsilon)\frac{1}{n}\mathbf{1} + \varepsilon\mathbf{a},$$

so that $A_0 = J/n$ and

$$\mathbf{s} = \sqrt{n} \left(\mathbf{a} - \frac{1}{n}\mathbf{1} \right).$$

Since $A'_0 \mathbf{s} = \mathbf{0}$, we have $\mathbf{g} = \mathbf{s}$. Moreover, every disagreement eigenvalue of A_0 is zero, so Proposition 4 gives

$$\mathbf{b}^*(\varepsilon) = \mathbf{u} + \varepsilon C \left(\mathbf{a} - \frac{1}{n}\mathbf{1} \right) + o(|\varepsilon|), \quad C > 0.$$

Thus the local response tilts toward leverage, recovering the ordering of Proposition 3.

¹⁰The reuse wedge vanishes as intervention becomes cheap: as $c \downarrow 0$, $M(\lambda) \rightarrow \lambda$. As $c \rightarrow \infty$, $M(\lambda) \rightarrow \lambda(1 - \delta)^2 / [(1 - \delta\lambda^2)(1 - \delta\lambda)]$.

5 Targeting flexibility

What is lost by fixing the exposure rule? We first characterize the value of retargeting, then identify environments in which one reusable direction is nearly sufficient even when exact replication requires more.

5.1 Value of flexibility

Recall the full-targeting benchmark from Section 2, under which the decision-maker chooses an unrestricted intervention each period, and let $v^f(\mathbf{x}_0)$ denote the full-targeting value. Define

$$\Delta(\mathbf{b}) = v^f(\mathbf{x}_0) - v(\mathbf{x}_0; \mathbf{b}) \geq 0, \quad \Delta^* = \min_{\|\mathbf{b}\|=1} \Delta(\mathbf{b}).$$

Since every coarse-targeting path is feasible under full targeting, $v^f(\mathbf{x}_0) \geq v(\mathbf{x}_0; \mathbf{b})$. Write the full-targeting value as $v^f(\mathbf{x}) = -\mathbf{x}' K^f \mathbf{x}$. The full and coarse intervention rules are

$$\mathbf{u}_t = -F^f \mathbf{x}_t, \quad F^f = \delta(cI + \delta K^f)^{-1} K^f A,$$

and

$$\mathbf{u}_t = -F^c(\mathbf{b}) \mathbf{x}_t, \quad F^c(\mathbf{b}) = \mathbf{b} \mathbf{p}(\mathbf{b})'.$$

The full-targeting problem assigns the quadratic weight $cI + \delta K^f$ to deviations from its optimal intervention. Define

$$\Phi(\mathbf{b}) = (F^c(\mathbf{b}) - F^f)'(cI + \delta K^f)(F^c(\mathbf{b}) - F^f).$$

Here $\Phi(\mathbf{b})$ measures the cost of intervention error, while $Z(\mathbf{b})$ records where the coarse path places discounted weight.

Proposition 5. *For any exposure rule satisfying $\|\mathbf{b}\|=1$,*

$$\Delta(\mathbf{b}) = \sum_{t=0}^{\infty} \delta^t \mathbf{x}_t' \Phi(\mathbf{b}) \mathbf{x}_t = \text{tr}(\Phi(\mathbf{b}) Z(\mathbf{b})),$$

where $\mathbf{x}_{t+1} = \tilde{A}(\mathbf{b}) \mathbf{x}_t$, $t \geq 0$.

A fixed one-dimensional rule can attain the full-targeting value even when its policy differs

from the full policy away from the realized path. Since $cI + \delta K^f$ is positive definite,

$$\Delta(\mathbf{b}) = 0 \iff (F^c(\mathbf{b}) - F^f)\mathbf{z} = \mathbf{0} \text{ for every } \mathbf{z} \in \text{Range}(Z(\mathbf{b})).$$

Thus a fixed rule need only reproduce full targeting on the path it induces. Because $Z(\mathbf{b})$ depends on $\mathbf{b}$, exact equivalence is an endogenous fixed-point condition. Writing

$$Z(\mathbf{b}) = \sum_{k=1}^n \sigma_k \mathbf{q}_k \mathbf{q}_k', \quad \sigma_1 \geq \dots \geq \sigma_n \geq 0,$$

gives

$$\Delta(\mathbf{b}) = \sum_{k=1}^n \sigma_k \mathbf{q}_k' \Phi(\mathbf{b}) \mathbf{q}_k.$$

Hence flexibility has little value when path weight is concentrated in a few directions and the policies agree there. We summarize path concentration by

$$d_{\text{eff}}(Z(\mathbf{b})) = \frac{\text{tr}(Z(\mathbf{b}))^2}{\text{tr}(Z(\mathbf{b})^2)}, \quad (9)$$

which equals one for a rank-one path and rises as path weight spreads across directions.¹¹

A simple bound follows when the network quickly eliminates inherited disagreement.

Corollary 1. *Suppose A is symmetric and $Q = I$, and write*

$$\mathbf{d} = \sum_{j>1} \tilde{x}_0^j \mathbf{p}_j.$$

Then

$$0 \leq \Delta^* \leq \sum_{j>1} \frac{\delta \lambda_j^2}{1 - \delta \lambda_j^2} (\tilde{x}_0^j)^2 \leq \frac{\delta |\lambda_2|^2}{1 - \delta |\lambda_2|^2} \|\mathbf{d}\|^2,$$

where $|\lambda_2| = \max_{j>1} |\lambda_j|$.

To see the bound, use the feasible uniform rule. Under symmetry and $Q = I$, it matches full

¹¹This differs from the one-dimensionality in Galeotti et al. (2020), which comes from concentration on a principal component of a static network game; here low dimensionality is a property of the endogenous path induced by the intervention rule.

targeting in the average-opinion dimension, while disagreement evolves as $A^t \mathbf{d}$. Hence

$$\Delta^* \leq \Delta(\mathbf{u}) \leq \sum_{t=1}^{\infty} \delta^t \|A^t \mathbf{d}\|^2 = \sum_{j>1} \frac{\delta \lambda_j^2}{1 - \delta \lambda_j^2} (\tilde{x}_0^j)^2,$$

and the final inequality follows from $|\lambda_j| \leq |\lambda_2|$.

Flexibility thus has little value when the disagreement actually present lies in directions the network quickly eliminates, even if other disagreement directions are persistent. The next subsection gives a different route: disagreement may remain persistent while additional targeting directions have little value.

5.2 When are simple instruments enough?

To locate exact benchmarks, first allow intervention to lie in an arbitrary fixed subspace $\mathcal{S} \subseteq \mathbb{R}^n$. Coarse targeting is the one-dimensional case $\mathcal{S} = \text{span}\{\mathbf{b}\}$, while full targeting corresponds to $\mathcal{S} = \mathbb{R}^n$.

Lemma 4. *A fixed subspace $\mathcal{S}$ attains the full-targeting value from $\mathbf{x}_0$ if and only if $F^f \mathbf{x}_t^f \in \mathcal{S}$ for every $t \geq 0$. Hence the minimum dimension required to reproduce full targeting is*

$$k_{\min} = \text{rank} \left(F^f Z^f F^{f'} \right), \quad Z^f = \sum_{t=0}^{\infty} \delta^t \mathbf{x}_t^f \mathbf{x}_t^{f'}.$$

To see this, note that

$$F^f Z^f F^{f'} = \sum_{t=0}^{\infty} \delta^t (F^f \mathbf{x}_t^f) (F^f \mathbf{x}_t^f)',$$

so its range is the span of the interventions used under full targeting. Thus $k_{\min}$ is the number of reusable directions required for exact replication. Because it is a rank condition, however, it counts every nonzero direction no matter how small its contribution. An arbitrarily small perturbation can therefore raise $k_{\min}$ even when the added directions have little value.

Example 1 (Exact dimension can jump). Consider $A(\rho)$ with $0 < \rho < 1$, $Q = I$, and an initial state with nonzero average and disagreement components. Because every disagreement direction has the same persistence ρ , the full-targeting intervention path lies in $\text{span}\{\mathbf{u}, \mathbf{d}\}$; with nonzero average and disagreement components, $k_{\min} = 2$. Now perturb

$A(\rho)$ by an arbitrarily small symmetric change that gives the $n - 1$ disagreement directions distinct persistence rates. For a generic initial state, their relative importance then changes over time and the full-targeting intervention path spans all n directions, so $k_{\min} = n$. Thus exact dimension can jump from 2 to n under an arbitrarily small change in the network (see Appendix S1 for a formal proof).

The next result shows that this fragility of exact dimension need not translate into a first-order loss in economic value. Consider the network family

$$A = \rho I + (1 - \rho)\mathbf{1}\mathbf{a}', \quad \mathbf{a} \geq \mathbf{0}, \quad \mathbf{a}'\mathbf{1} = 1, \quad \rho \in [0, 1).$$

Every agent places the same relative weight $\mathbf{a}$ on others, while retaining common self-weight ρ . The network may therefore have unequal leverage even though every disagreement direction is propagated at the same rate. Recall that $\Pi = I - \mathbf{u}\mathbf{u}'$ projects onto the disagreement space and $A_{\perp} = \Pi A \Pi$ describes the propagation of disagreement:

$$A_{\perp} = \rho \Pi.$$

The family includes $A(\rho)$ when $\mathbf{a} = \mathbf{1}/n$ and the opinion-leader network when $\rho = 0$. To isolate the disagreement problem, starting from centered disagreement $\mathbf{d} \neq \mathbf{0}$,

$$A\mathbf{d} = \rho\mathbf{d} + \mathbf{u}(\ell'\mathbf{d}).$$

If $\ell'\mathbf{d} = 0$, full targeting remains on $\text{span}\{\mathbf{d}\}$, so $k_{\min} \leq 1$. Even when $\ell'\mathbf{d} \neq 0$, the full-targeting intervention path remains in $\text{span}\{\mathbf{u}, \mathbf{d}, \ell\}$, so $k_{\min} \leq 3$ regardless of n .

We therefore perturb the two conditions behind exact one-dimensionality. Measure departures from common persistence by $A_{\perp} - \rho\Pi$ and leverage–disagreement alignment by $|\ell'\mathbf{d}|/\|\mathbf{d}\|$. We use the Euclidean norm for vectors and the Frobenius norm for matrices.

Proposition 6. *Suppose $Q = I$ and consider the problem starting from $\mathbf{d} \neq \mathbf{0}$. Fix a benchmark network A_0 with common persistence parameter ρ , so that*

$$A_{0,\perp} = \rho\Pi, \quad \ell'_0\mathbf{d} = 0.$$

There exist $\eta > 0$ and $C > 0$ such that every row-stochastic network A satisfying $\|A - A_0\| <$

η obeys

$$\Delta^* \leq C \left(\|\mathbf{d}\|^2 \|A_\perp - \rho \Pi\|^2 + (\ell' \mathbf{d})^2 \right).$$

The perturbation need not preserve symmetry.

Exact dimension and economic value can therefore come apart: exact replication of full targeting may require many additional directions after an arbitrarily small perturbation, while the value lost from retaining one reusable direction is only second order in the perturbation.

Proposition 6 complements Corollary 1. There flexibility has little value because disagreement fades quickly. Here disagreement may remain highly persistent, yet one reusable direction is nearly sufficient when different forms of disagreement are propagated similarly and leverage is weakly aligned with the disagreement present. These conditions are sufficient rather than exhaustive. The application next allows asymmetric propagation and population-weighted losses.

6 Empirical illustration: climate-opinion disagreement

We illustrate the theory using U.S. state-level climate opinions and the Facebook Social Connectedness Index (SCI). We do not estimate opinion dynamics; we use observed disagreement and social connectedness to compute the exposure rule and the value of targeting flexibility implied by the model. The application does not impose the sufficient conditions of Proposition 6; instead, we use the general value decomposition in Proposition 5.

6.1 Data and construction

We construct state-level initial opinions from the 2023 Yale Climate Opinion Maps (Howe et al., 2015; Marlon et al., 2022), population-weighting county estimates within each state. The initial opinion $x_{0,s}$ is the estimated share of residents in state s who disagree with the statement “Most scientists think global warming is happening.” We interpret this as disagreement with the scientific-consensus benchmark and normalize the decision-maker’s agenda to zero. Figure 1 shows the resulting opinion vector.

We measure network links using the Facebook SCI, aggregated from counties to states

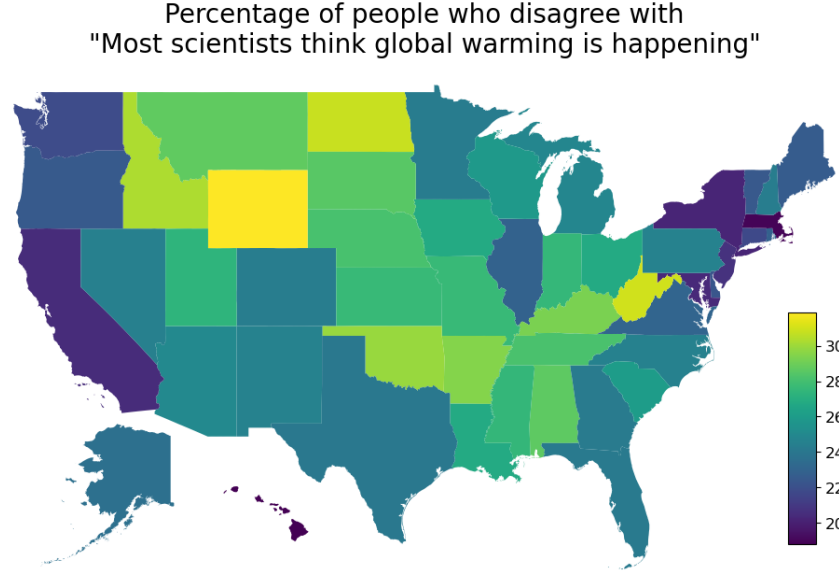


Figure 1: Disagreement with the scientific consensus across U.S. states.

following Bailey et al. (2021). The construction derived in Appendix S5 gives

$$\hat{A}_{ss'} = \frac{\text{SCI}_{ss'} N_{s'}}{\sum_{s''} \text{SCI}_{ss''} N_{s''}},$$

where $N_{s'}$ is the population of destination state s' . Thus $\hat{A}_{ss'}$ is the weight state s places on opinions in state s' . The resulting row-stochastic network is asymmetric and geographically clustered (see Figure 5 in Appendix S5). Authority and future persistence therefore need not point toward the same states: a state may be influential in the propagation network, or it may be connected to disagreement that the network continues to carry forward.

6.2 Which states receive exposure?

To connect the application to the two forces in Section 4, we compare the reusable rule with measures of authority and future persistence. Authority is not identical to leverage in a general asymmetric network, but serves here as a familiar empirical summary of the influence side of the theory.

We compute the highest-value stationary coarse rule found by the multi-start procedure in Appendix S5.2, denoted $\hat{\mathbf{b}}$. Our baseline uses $\delta = 0.9$, $c = 10$, and population-weighted

losses,

$$Q^{\text{pop}} = \text{diag}(N_s/\bar{N}),$$

where $\bar{N}$ is average state population. These parameters are illustrative rather than calibrated; one period denotes one application of $\hat{A}$.

We use the leading right singular vector of $\hat{A}$, denoted by $\mathbf{v}_1$ and normalized to unit Euclidean norm, as the authority benchmark.¹² To summarize persistence, let $\mathbf{d} = \mathbf{x}_0 - \bar{x}_0 \mathbf{1}$ denote centered cross-state disagreement and define

$$\mathbf{h}^P = \hat{A}(I - \delta \hat{A})^{-1} \mathbf{d} = \hat{A} \mathbf{d} + \delta \hat{A}^2 \mathbf{d} + \dots$$

A higher h_s^P means that current cross-state disagreement contributes more positively to state s 's future opinion path. Because the score is signed, a higher value may simply be less negative.

Figure 2 compares the reusable exposure rule with authority and future persistence across states. California has substantially greater authority than Texas, while Texas has the higher future-persistence score. Despite an authority gap of 0.071, their exposure weights differ by only 0.003. Ohio provides the complementary comparison: its authority is much lower, but its higher future-persistence score goes with exposure substantially above its authority weight.

Comparing the reusable rule with static exposure isolates the effect of reuse. Static exposure is

$$\mathbf{b}^{\text{static}} \propto (cI + \delta Q^{\text{pop}})^{-1} Q^{\text{pop}} \hat{A} \mathbf{x}_0.$$

At the baseline, reuse lowers California's exposure from 0.437 to 0.328 and Texas's from 0.398 to 0.325.¹³ Thus reuse materially reallocates exposure even though, as the next subsection shows, these reallocations have little effect on value. Table 1 summarizes the baseline comparisons.

¹²The ordering of these three states is unchanged if authority is instead measured by the invariant distribution or column sums of $\hat{A}$. California remains above Texas, and Ohio remains below both.

¹³Appendix Table 5 reports the largest state-level reallocations, including increases for Mississippi, Minnesota, and Alabama.

Table 1: Authority, future persistence, and exposure at the baseline ($\delta = 0.9$, $c = 10$). Authority, future persistence, reusable exposure, and static exposure are normalized to unit Euclidean norm; future persistence is signed. The final row reports the difference between the two exposure directions.

	Texas	California	Ohio
Authority	0.3430	0.4140	0.1600
Future persistence	-0.0760	-0.1500	-0.0230
Reusable exposure	0.3248	0.3280	0.2195
Static exposure	0.3983	0.4366	0.2096
Reusable minus static	-0.0735	-0.1086	0.0099

6.3 How valuable is targeting flexibility?

We compare the reported coarse rule with full targeting, fixed exposure directions, and period-by-period myopic retargeting.¹⁴ Fixed exposure directions, including static exposure, use their optimal dynamic scalar policies, while the myopic policy reoptimizes an unrestricted one-period vector intervention each period. For a rule with value v , define its share of attainable gains as

$$\frac{v - v^\emptyset}{v^f - v^\emptyset}.$$

Table 2 shows that across the displayed intervention costs, the reported coarse rule captures

Table 2: Share of attainable gains captured ($\delta = 0.9$, $Q = Q^{\text{pop}}$). Entries are rounded to the nearest tenth of a percentage point.

	$c = 1$	$c = 10$	$c = 100$
Opinion ($\mathbf{b} \propto \mathbf{x}_0$)	98.3%	91.1%	72.0%
Authority ($\mathbf{b} \propto \mathbf{v}_1$)	97.2%	99.1%	97.8%
Static exposure	99.6%	99.2%	99.5%
Myopic retargeting	99.7%	91.5%	52.2%
Coarse ($\mathbf{b} = \hat{\mathbf{b}}$)	99.9%	99.9%	100.0%

at least 99.9% of attainable gains. This is not because policy choice is uniformly unimportant: at $c = 100$, the opinion direction captures only 72.0% of attainable gains and period-by-period myopic retargeting only 52.2%, while the reusable rule remains essentially at the full-targeting value. Flexibility does not compensate for ignoring continuation values. Static

¹⁴Because $\hat{\mathbf{b}}$ is feasible, its reported capture is a lower bound on optimal coarse capture.

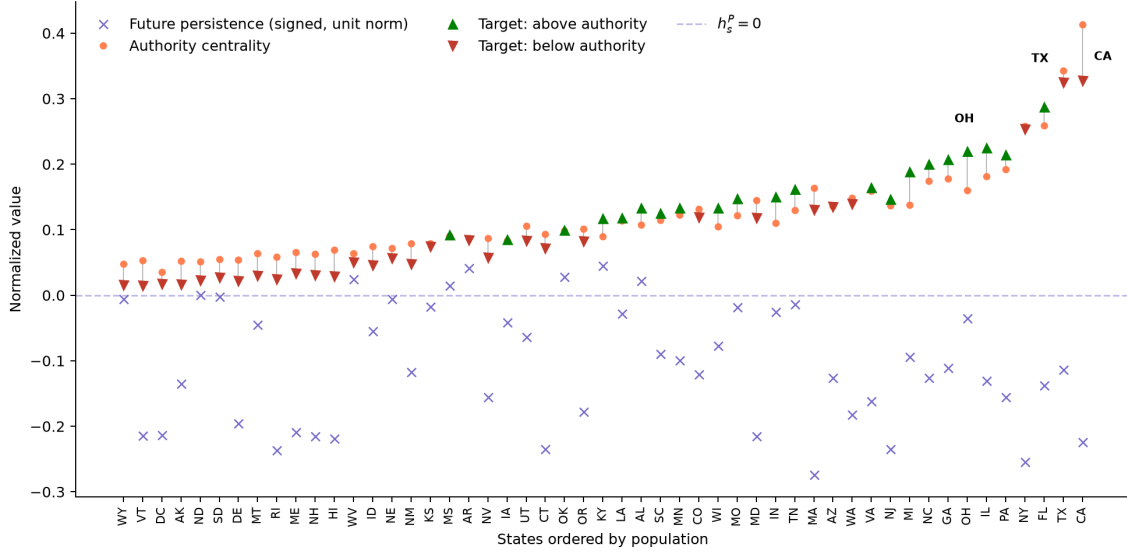


Figure 2: Population-weighted exposure compared with authority and future persistence across states. Authority and exposure have unit Euclidean norm; future persistence is signed and normalized to unit Euclidean norm.

exposure also remains close to the reusable rule in value despite the substantial reallocation under reuse documented in Section 6.2.

Proposition 5 gives the mechanism:

$$\Delta(\hat{\mathbf{b}}) = \text{tr}\left(\Phi(\hat{\mathbf{b}})Z(\hat{\mathbf{b}})\right).$$

At the baseline, the realized path has effective dimension 1.03, with 98.59% of discounted path weight in its leading direction. Yet that direction contributes only 1.14% of the total coarse-to-full gap: the interventions prescribed by the coarse and full policies there have cosine 0.99998. Their larger discrepancies therefore lie mainly in directions that the realized path scarcely visits.

What produces this concentration? Two natural explanations can be ruled out. First, it is not driven by the large common opinion component. Although the common component accounts for 98.0% of $\|\mathbf{x}_0\|^2$, replacing $\mathbf{x}_0$ by centered disagreement $\mathbf{d}$ leaves the path highly concentrated: 96.24% of discounted path weight remains in one direction, and the reported reusable rule still captures 99.61% of attainable gains.

Second, the concentration is not a special alignment between the Facebook network and

Table 3: Baseline and centered-disagreement paths ($\delta = 0.9$, $c = 10$, $Q = Q^{\text{pop}}$).

Initial opinion	Coarse capture	$d_{\text{eff}}(Z)$	Lead share
$\mathbf{x}_0$	99.91%	1.03	98.59%
$\mathbf{d}$	99.61%	1.08	96.24%

the observed placement of disagreement. Randomly reshuffling disagreement across states leaves the Facebook path nearly one-dimensional: the median leading share remains 98.0% and the median effective dimension is 1.04. By contrast, random row-stochastic networks calibrated to have a similar median second-eigenvalue modulus yield a median leading share of only 63.8% and a median effective dimension of 2.16; Appendix Table 9 reports the full comparison. These exercises point to the structure of the Facebook propagation matrix, rather than the placement of disagreement or persistence as summarized by the second-eigenvalue modulus, as the source of the concentration.

Recurrent shocks provide a boundary. The cross-state standard deviation of initial disagreement is about 0.036, so a shock standard deviation of $\sigma = 0.1$ is almost three times that dispersion. Coarse targeting still captures 97.2% of attainable gains at that shock size, but loses value as shocks populate new directions.

7 The dimension of new disagreement

Suppose iid agent-level shocks arrive after intervention:

$$\mathbf{x}_{t+1} = A\mathbf{x}_t + \mathbf{u}_t + \boldsymbol{\nu}_t, \quad \boldsymbol{\nu}_t \sim \mathcal{N}(\mathbf{0}, \sigma^2 I).$$

The decision-maker observes $\mathbf{x}_t$ before intervening. For symmetric $A = PDP'$, write $\tilde{\mathbf{x}}_t = P'\mathbf{x}_t$ and $\tilde{\boldsymbol{\nu}}_t = P'\boldsymbol{\nu}_t$, so $\tilde{\boldsymbol{\nu}}_t \sim \mathcal{N}(\mathbf{0}, \sigma^2 I)$.

We compare full targeting, coarse targeting with $\tilde{\mathbf{u}}_t = \tilde{\mathbf{b}}r_t$, and myopic targeting, which retains unrestricted intervention but treats the next period as terminal:

$$\tilde{\mathbf{u}}_t^m = -\frac{\delta}{c + \delta} D\tilde{\mathbf{x}}_t.$$

Myopic targeting preserves dimensional flexibility but ignores the persistence of the disagreement it moves.

Because the shocks are additive and mean zero, they do not change the feedback response to current opinions conditional on an exposure rule; they affect the choice of that rule through the additional expected loss from recurrent shocks. For each rule $q \in \{f, m, c\}$,

$$\tilde{v}^q(\tilde{\mathbf{x}}_0) = -\tilde{\mathbf{x}}_0' \tilde{K}^q \tilde{\mathbf{x}}_0 - \frac{\delta \sigma^2}{1 - \delta} \text{tr}(\tilde{K}^q),$$

where the two terms capture losses from initial opinions and recurrent shocks, respectively.

Proposition 7. *Suppose A is symmetric, $Q = I$, $\boldsymbol{\nu}_t \sim \mathcal{N}(\mathbf{0}, \sigma^2 I)$, and $\sigma > 0$.*

- (1) *Full targeting strictly dominates myopic targeting.*
- (2) *Full targeting weakly dominates coarse targeting, with equality if $\text{rank}(A) = 1$ and strict dominance if $\text{rank}(A) > 1$.*

The comparison separates foresight from dimensional flexibility. Proposition 8 shows which advantage dominates when recurrent shocks continually create new disagreement in the highly symmetric family in (8).

Proposition 8. *Suppose $Q = I$, $A = A(\rho)$, $\sigma > 0$ and $\mathbf{d} \neq \mathbf{0}$.*

- (1) *For each fixed n , an optimal coarse exposure rule can be chosen in $\text{span}\{\mathbf{u}, \mathbf{d}\}$.*
- (2) *Fix $\rho \in (0, 1]$ and consider a sequence of such networks as $n \rightarrow \infty$. If $|\mathbf{u}'_n \mathbf{x}_0^{(n)}| = O(1)$ and $\|\mathbf{d}_n\| = O(1)$, then*

$$1 > \lim_{n \rightarrow \infty} \frac{\tilde{v}_n^f}{\tilde{v}_n^m} > \lim_{n \rightarrow \infty} \frac{\tilde{v}_n^f}{\tilde{v}_n^c} > 0.$$

Part (1) shows that the span restriction from Section 4.1 survives recurrent shocks: because all disagreement directions orthogonal to $\mathbf{d}$ are ex ante symmetric, an optimal fixed rule need not select among them. Realized shocks nevertheless populate all of these directions.

Part (2) delivers the reversal. As the network grows, myopic targeting approaches full targeting more closely than coarse targeting does: dimensional flexibility outweighs foresight because a fixed rule omits an increasing number of realized disagreement directions.

In the same $A(\rho)$ sequence, if an optimally chosen fixed subspace has dimension k_n with $k_n/n \rightarrow \alpha$, it captures asymptotically an α share of the gains from controlling recurrent disagreement relative to no intervention. Preserving a fixed share of those gains therefore

requires the number of reusable directions to grow proportionally with the number of disagreement dimensions.

The polar cases make the mechanism transparent. In the clique ($\rho = 0$), disagreement disappears after one update and coarse targeting matches full targeting. In the identity network, the deterministic path is concentrated, but recurrent shocks populate every disagreement direction, so a fixed rule loses value relative to full targeting as σ increases (Figure 3).

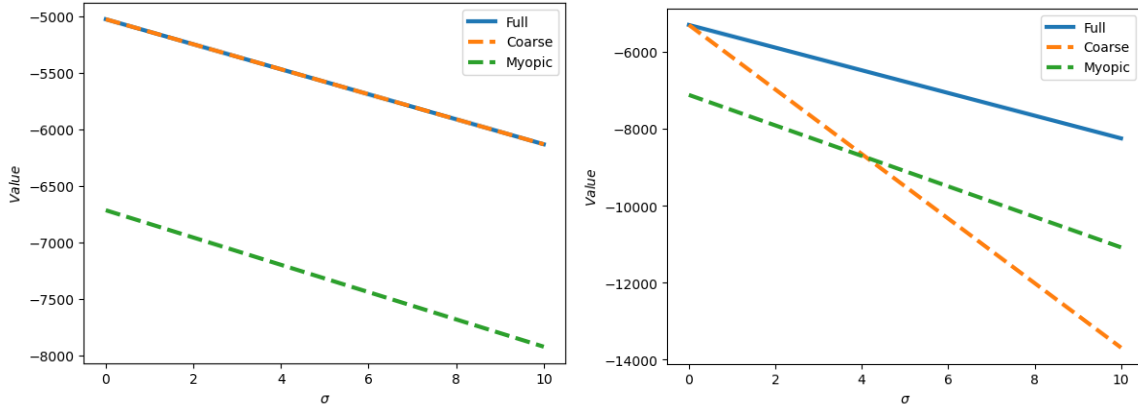


Figure 3: Value comparison across targeting methods in the clique (left) and the identity network (right) as the scale of recurrent shocks increases.

A predictable bias is different: it follows the predetermined orbit $\mathbf{w}, A\mathbf{w}, A^2\mathbf{w}, \dots$, so recurrence need not make the relevant policy space diffuse, as shown in Appendix S2.

8 Conclusion

High-dimensional targeting need not require a high-dimensional policy. When an exposure rule must be reused, what matters is not how many directions are available, but which ones the network keeps economically relevant.

This distinction can be sharp. Exact replication of full targeting may require many intervention directions, while one reusable rule captures nearly all of their value. The reason is that policy flexibility matters only where the endogenous opinion path places weight. Persistent disagreement can therefore remain simple to target; what defeats a simple instrument is the continual arrival of new disagreement in many directions.

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A Proofs

Standard or subsidiary results and proofs are relegated to Section S1 of the Supplementary Appendix. They include Lemma 1, Lemma 2, Corollary S1, Lemma 3, Lemma 5 and Lemma 4, with the remaining supporting results proved below.

A.1 Proofs from Section 3

Proof of Proposition 1: We derive the gradient of the value with respect to the exposure rule. Fix $\mathbf{b} \neq \mathbf{0}$. By Lemma 1, the optimal intervention level is unique and linear: there exists a vector $\mathbf{p}$ such that $r_t = -\mathbf{p}'\mathbf{x}_t$ for all $t \geq 0$. Under this policy, $\mathbf{x}_{t+1} = \tilde{A}(\mathbf{b})\mathbf{x}_t$ where $\tilde{A}(\mathbf{b}) \equiv A - \mathbf{b}\mathbf{p}'$. By Lemma 1, $\rho(\sqrt{\delta} \tilde{A}(\mathbf{b})) < 1$, so all discounted sums below are well defined. For any opinion $\mathbf{x}_t$, the Bellman equation evaluated at the optimal r_t is

$$v(\mathbf{x}_t; \mathbf{b}) = -\mathbf{x}_t' Q \mathbf{x}_t - c r_t^2 - \delta \mathbf{x}_{t+1}' K^*(\mathbf{b}) \mathbf{x}_{t+1}, \text{ and } \mathbf{x}_{t+1} = A \mathbf{x}_t + \mathbf{b} r_t.$$

By the Envelope theorem, terms involving $\partial r_t / \partial \mathbf{b}$ vanish. Holding the current opinion fixed, $\partial \mathbf{x}_{t+1} / \partial \mathbf{b} = r_t I$. Differentiating with respect to $\mathbf{b}$,

$$\nabla_{\mathbf{b}} v(\mathbf{x}_t; \mathbf{b}) = -2\delta r_t K^*(\mathbf{b}) \mathbf{x}_{t+1} + \delta \nabla_{\mathbf{b}} v(\mathbf{x}_{t+1}; \mathbf{b}).$$

Iterating forward from $t = 0$ and using $\rho(\sqrt{\delta} \tilde{A}(\mathbf{b})) < 1$ gives

$$\nabla_{\mathbf{b}} v(\mathbf{x}_0; \mathbf{b}) = -2\delta \sum_{t=0}^{\infty} \delta^t r_t K^*(\mathbf{b}) \mathbf{x}_{t+1}.$$

Substituting $r_t = -\mathbf{p}'\tilde{A}(\mathbf{b})^t\mathbf{x}_0$ and $\mathbf{x}_{t+1} = \tilde{A}(\mathbf{b})^{t+1}\mathbf{x}_0$ yields

$$\nabla_{\mathbf{b}}v(\mathbf{x}_0; \mathbf{b}) = 2\delta K^*(\mathbf{b})\tilde{A}(\mathbf{b}) \left(\sum_{t=0}^{\infty} (\sqrt{\delta} \tilde{A}(\mathbf{b}))^t \mathbf{x}_0 \mathbf{x}_0' (\sqrt{\delta} \tilde{A}(\mathbf{b})')^t \right) \mathbf{p}.$$

Recall that $Z \equiv \sum_{t=0}^{\infty} (\sqrt{\delta} \tilde{A}(\mathbf{b}))^t \mathbf{x}_0 \mathbf{x}_0' (\sqrt{\delta} \tilde{A}(\mathbf{b})')^t$. Then $\nabla_{\mathbf{b}}v(\mathbf{x}_0; \mathbf{b}) = 2\delta K^*(\mathbf{b})\tilde{A}(\mathbf{b})Z\mathbf{p}$. Moreover, because $\rho(\sqrt{\delta} \tilde{A}(\mathbf{b})) < 1$, Z is the unique solution to the Lyapunov equation

$$Z = \delta \tilde{A}(\mathbf{b})Z\tilde{A}(\mathbf{b})' + \mathbf{x}_0\mathbf{x}_0'.$$

The Lagrangian for the constrained problem on the unit sphere is

$$\mathcal{L}(\mathbf{b}, \lambda) = v(\mathbf{x}_0; \mathbf{b}) - \lambda(\mathbf{b}'\mathbf{b} - 1).$$

At a constrained stationary point on the sphere, $\nabla_{\mathbf{b}}v(\mathbf{x}_0; \mathbf{b}) = 2\lambda\mathbf{b}$. Using the gradient expression above and absorbing the positive scalar 2δ into the multiplier gives $\mu\mathbf{b} = K^*(\mathbf{b})\tilde{A}(\mathbf{b})Z\mathbf{p}$, which is (7). At $\mathbf{b} = \mathbf{0}$, define

$$K^*(\mathbf{0}) = \sum_{t=0}^{\infty} \delta^t (A')^t Q A^t,$$

the no-intervention loss matrix. The fixed-rule value extends continuously to $\mathbf{b} = \mathbf{0}$. Hence $v(\mathbf{x}_0; \mathbf{b})$ is continuous on the compact feasible set $\{\mathbf{b} : \|\mathbf{b}\| \leq 1\}$. Existence follows from continuity of $v(\mathbf{x}_0; \mathbf{b})$ and compactness of the feasible set $\{\mathbf{b} : \|\mathbf{b}\| \leq 1\}$. If $Q A \mathbf{x}_0 \neq \mathbf{0}$, Lemma 2 implies that every optimizer satisfies $\|\mathbf{b}^*\| = 1$. Hence every optimizer satisfies the first-order condition above. $\square$

A.2 Proofs from Section 4

Proof of Proposition 2: Let $\Pi = I - \mathbf{u}\mathbf{u}'$ denote the projection onto deviations from average opinion. Decompose any exposure rule as $\mathbf{b} = \alpha\mathbf{u} + \mathbf{q}$, $\alpha = \mathbf{u}'\mathbf{b}$, and $\mathbf{q} \perp \mathbf{u}$. Recall $\bar{x}_t = \frac{1}{n}\mathbf{1}'\mathbf{x}_t = \frac{1}{\sqrt{n}}\mathbf{u}'\mathbf{x}_t$. Because A is doubly stochastic, $A\mathbf{u} = \mathbf{u}$ and $\mathbf{u}'A = \mathbf{u}'$. Therefore, under $(\mathbf{b}, \{r_t\})$, average opinion evolves according to

$$\bar{x}_{t+1} = \bar{x}_t + \frac{\alpha}{\sqrt{n}}r_t.$$

Suppose first that $Ad = \mathbf{0}$. Given any feasible exposure rule $\mathbf{b} = \alpha\mathbf{u} + \mathbf{q}$ and intervention sequence $\{r_t\}$, consider uniform exposure $\mathbf{u}$ with $\hat{r}_t = \alpha r_t$. Thus $\bar{x}_{t+1} = \bar{x}_t + \frac{1}{\sqrt{n}}\hat{r}_t$. The two policies generate the same path of average opinion. Moreover, $Ad = \mathbf{0}$, so inherited disagreement disappears after the first update. Uniform exposure creates no new disagreement, whereas a nonuniform exposure rule may do so. Since

$$\|\mathbf{x}_t\|^2 = n\bar{x}_t^2 + \|\Pi\mathbf{x}_t\|^2,$$

uniform exposure therefore generates weakly lower opinion loss at every date. Its intervention cost is also weakly lower because $\hat{r}_t^2 = \alpha^2 r_t^2 \leq r_t^2$, where $|\alpha| \leq \|\mathbf{b}\| \leq 1$. Thus uniform targeting weakly dominates every exposure rule and is optimal.

Conversely, suppose that $Ad \neq \mathbf{0}$ and consider uniform exposure, $\mathbf{b} = \mathbf{u}$. Since both $\text{span}\{\mathbf{u}\}$ and $\mathbf{u}^\perp$ are invariant under A and A' , the Riccati map preserves matrices that do not mix the average-opinion direction and the disagreement space. By uniqueness of the stabilizing Riccati solution,

$$K^* = \tilde{k}_1 \mathbf{u}\mathbf{u}' + K_\perp, \quad K_\perp = \Pi K^* \Pi,$$

where $K_\perp$ is positive definite on $\mathbf{u}^\perp$. Moreover,

$$\mathbf{p} = p_1 \mathbf{u}, \quad \tilde{A}(\mathbf{u}) = a_1 \mathbf{u}\mathbf{u}' + A_\perp, \quad A_\perp = \Pi A \Pi,$$

where

$$p_1 = \frac{\delta \tilde{k}_1}{c + \delta \tilde{k}_1} > 0, \quad a_1 = 1 - p_1 = \frac{c}{c + \delta \tilde{k}_1} \in (0, 1).$$

Recall $\bar{x}_0 = \frac{1}{n} \mathbf{1}' \mathbf{x}_0$, $\mathbf{d} = \Pi \mathbf{x}_0$ so that $\mathbf{u}' \mathbf{x}_0 = \sqrt{n} \bar{x}_0$. Since

$$\tilde{A}(\mathbf{u}) = a_1 \mathbf{u}\mathbf{u}' + A_\perp,$$

we have

$$\Pi \tilde{A}(\mathbf{u})^t \mathbf{x}_0 = A_\perp^t \mathbf{d}, \quad \mathbf{x}_0' (\tilde{A}(\mathbf{u})')^t \mathbf{u} = a_1^t \sqrt{n} \bar{x}_0.$$

Hence

$$\Pi Z \mathbf{u} = \sqrt{n} \bar{x}_0 (I - \delta a_1 A_\perp)^{-1} \mathbf{d}.$$

Because A is doubly stochastic, $\|A\|_2 \leq \sqrt{\|A\|_1 \|A\|_\infty} = 1$. Hence $\rho(A_\perp) \leq \|A_\perp\|_2 \leq 1$.

Since $\delta a_1 < 1$, the matrix $I - \delta a_1 A_\perp$ is invertible. If uniform exposure were optimal, Proposition 1 would imply that it satisfies (7). Projecting the right-hand side of (7) onto $\mathbf{u}^\perp$ gives

$$\Pi K^* \tilde{A}(\mathbf{u}) Z \mathbf{p} = \sqrt{n} \bar{x}_0 p_1 K_\perp A_\perp (I - \delta a_1 A_\perp)^{-1} \mathbf{d}.$$

Since $K_\perp$ and $I - \delta a_1 A_\perp$ are invertible on $\mathbf{u}^\perp$, this expression is zero if and only if $A_\perp \mathbf{d} = A \mathbf{d} = \mathbf{0}$. Thus when $A \mathbf{d} \neq \mathbf{0}$, its right-hand side has a nonzero disagreement component, whereas $\mathbf{b} = \mathbf{u}$ has none. Uniform exposure therefore cannot be optimal. $\square$

Proof of Proposition 3: By Lemma 2, an optimal exposure rule has $\|\mathbf{b}\| = 1$. Fix such a rule. Since $A = \mathbf{1} \mathbf{a}'$, the Riccati equation implies

$$K = I + \nu \mathbf{a} \mathbf{a}'$$

for some $\nu > 0$. Hence

$$\mathbf{x}_0' K \mathbf{x}_0 = \|\mathbf{x}_0\|^2 + \nu (\mathbf{a}' \mathbf{x}_0)^2.$$

For every $\mathbf{x}_0$ with $\mathbf{a}' \mathbf{x}_0 \neq 0$, the optimal exposure rule therefore minimizes the same scalar ν and is independent of $\mathbf{x}_0$. Substituting $K = I + \nu \mathbf{a} \mathbf{a}'$ into the Riccati equation gives

$$(1 - \delta) \nu - \delta n + \delta^2 R(\mathbf{b}, \nu) = 0,$$

where

$$R(\mathbf{b}, \nu) = \frac{((\mathbf{1} + \nu \mathbf{a})' \mathbf{b})^2}{\mathbf{b}' ((c + \delta) I + \delta \nu \mathbf{a} \mathbf{a}') \mathbf{b}}.$$

At an optimum, the directional derivative of $\nu(\mathbf{b})$ vanishes along every tangent direction to the unit sphere. Differentiating the equation above therefore implies that $\mathbf{b}^*$ is a constrained critical point of $R(\mathbf{b}, \nu^*)$.

For any positive definite M and nonzero $\mathbf{y}$, the nonzero constrained critical points of $\frac{(\mathbf{y}' \mathbf{b})^2}{\mathbf{b}' M \mathbf{b}}$ are proportional to $M^{-1} \mathbf{y}$. The alternative $\mathbf{y}' \mathbf{b} = 0$ gives identically zero intervention and cannot be optimal when $\mathbf{a}' \mathbf{x}_0 \neq 0$. Hence, with

$$M = (c + \delta) I + \delta \nu \mathbf{a} \mathbf{a}', \quad \mathbf{y} = \mathbf{1} + \nu \mathbf{a},$$

we have

$$\mathbf{b}^* \propto ((c + \delta) I + \delta \nu \mathbf{a} \mathbf{a}')^{-1} (\mathbf{1} + \nu \mathbf{a}).$$

By the Sherman–Morrison formula,

$$\mathbf{b}^* \propto \mathbf{1} + q\mathbf{a}, \quad q = \frac{\nu c}{c + \delta + \delta\nu\|\mathbf{a}\|^2} > 0.$$

Since

$$\mathbf{a} = \frac{\mathbf{1}}{n} + \frac{\boldsymbol{\ell}}{\sqrt{n}},$$

we have

$$\mathbf{1} + q\mathbf{a} = \sqrt{n} \left(1 + \frac{q}{n}\right) \mathbf{u} + \frac{q}{\sqrt{n}} \boldsymbol{\ell}.$$

Hence, under the normalization $\mathbf{a}'\mathbf{b}^* > 0$,

$$\mathbf{b}^* = \alpha\mathbf{u} + \gamma\boldsymbol{\ell}, \quad \alpha > \gamma > 0.$$

Moreover, $b_i^* - b_j^* = \gamma(\ell_i - \ell_j) = \gamma\sqrt{n}(a_i - a_j)$, and $b_i^* = \frac{\alpha-\gamma}{\sqrt{n}} + \gamma\sqrt{n}a_i > 0$. Therefore $b_1^* > \dots > b_n^* > 0$. $\square$

Proof of Proposition 4: Work in the eigenbasis of A_0 and let $P = (\mathbf{p}_1, \dots, \mathbf{p}_n)$ where $\mathbf{p}_1 = \mathbf{u}$. Thus $\mathbf{e}_j$ denotes the j th coordinate vector in the transformed basis. Let $\tilde{\tilde{A}} = P'\tilde{A}P$ denote the induced opinion-transition matrix in the eigenbasis. Then $\tilde{\tilde{A}}_0 = \text{diag}(\tau, \lambda_2, \dots, \lambda_n)$. By scale invariance, any initial opinion near $\mathbf{x}_0 \propto \mathbf{1}$ can be normalized as

$$\tilde{\mathbf{x}}_0 = \mathbf{e}_1 + \boldsymbol{\xi}, \quad \boldsymbol{\xi} \perp \mathbf{e}_1.$$

Parameterize unit exposure rules near $\mathbf{e}_1$ by

$$\tilde{\mathbf{b}}(\boldsymbol{\zeta}) = \sqrt{1 - \|\boldsymbol{\zeta}\|^2} \mathbf{e}_1 + \boldsymbol{\zeta}, \quad \boldsymbol{\zeta} \perp \mathbf{e}_1.$$

Hence

$$\tilde{\mathbf{b}}(\boldsymbol{\zeta}) = \mathbf{e}_1 + \boldsymbol{\zeta} + O(\|\boldsymbol{\zeta}\|^2).$$

At the benchmark $(\boldsymbol{\zeta}, \boldsymbol{\xi}, \varepsilon) = (\mathbf{0}, \mathbf{0}, 0)$, the Riccati matrix $\tilde{K}_0 = \text{diag}(\tilde{k}_1, \dots, \tilde{k}_n)$. The closed-loop matrix is $\tilde{\tilde{A}}_0 = \text{diag}(\tau, \lambda_2, \dots, \lambda_n)$. Moreover, $\tilde{\mathbf{p}}_0 = \ell_1 \mathbf{e}_1$ where

$$\ell_1 = \frac{\delta\tilde{k}_1}{c + \delta\tilde{k}_1}, \quad \tau = 1 - \ell_1 = \frac{c}{c + \delta\tilde{k}_1} \in (0, 1),$$

and, for $j > 1$, $\tilde{k}_j = \frac{1}{1-\delta\lambda_j^2}$. Since (δ, c) are fixed, the average-coordinate equations determine τ , $\tilde{k}_1$, and ℓ_1 . Hence the coefficients below depend on j only through λ_j . Also,

$$\tilde{Z}_0 = z_1 \mathbf{e}_1 \mathbf{e}_1', \quad z_1 = \frac{1}{1 - \delta\tau^2}.$$

We first compute the tangent Hessian common to both perturbations. For each $j > 1$, consider the one-dimensional exposure rotation

$$\tilde{\mathbf{b}}_j(\gamma) = \sqrt{1 - \gamma^2} \mathbf{e}_1 + \gamma \mathbf{e}_j$$

and write $\tilde{K}_{1j}(\gamma) = q_j \gamma + o(\gamma)$ and $\tilde{K}_{11}(\gamma) = \tilde{k}_1 + h_j \gamma^2 + o(\gamma^2)$. Matching the coefficient of γ in the $(1, j)$ entry of the transformed Riccati equation gives q_j ; substituting this into the $(1, 1)$ entry and matching the coefficient of γ^2 gives h_j . Thus

$$q_j = -\frac{\delta\lambda_j\ell_1\tilde{k}_j}{1 - \delta\tau\lambda_j} \quad \text{and} \quad h_j = \frac{\delta \left[\ell_1\tau(\tilde{k}_1 - 2q_j) + \ell_1^2\tilde{k}_j \right]}{1 - \delta\tau^2}.$$

Because $|\lambda_j| \leq 1$,

$$1 - \delta\tau\lambda_j > 0, \quad 1 + \delta\tau\lambda_j > 0,$$

and

$$\tau(\tilde{k}_1 - 2q_j) + \ell_1\tilde{k}_j = \frac{\tau\tilde{k}_1(1 - \delta\tau\lambda_j) + \ell_1\tilde{k}_j(1 + \delta\tau\lambda_j)}{1 - \delta\tau\lambda_j} > 0.$$

Hence $h_j > 0$ for every $j > 1$.

At the benchmark, reversing the sign of any disagreement coordinate leaves D , $\mathbf{e}_1$, and $Q = I$ unchanged. Hence value is invariant under $\zeta_j \mapsto -\zeta_j$ for every $j > 1$. It is therefore even in each ζ_j , which rules out all quadratic cross terms $\zeta_j\zeta_l$, $j \neq l$. The constrained Hessian of value with respect to $\boldsymbol{\zeta}$ is consequently $-2 \text{diag}(h_2, \dots, h_n)$. Because $h_j > 0$ for every $j > 1$, this Hessian is negative definite, so $\boldsymbol{\zeta} = \mathbf{0}$ is a nondegenerate strict local maximum at the benchmark. Let

$$\Psi(\boldsymbol{\zeta}; \boldsymbol{\xi}, \varepsilon) = D_{\boldsymbol{\zeta}} \tilde{v}(\mathbf{e}_1 + \boldsymbol{\xi}; \tilde{\mathbf{b}}(\boldsymbol{\zeta}), D + \varepsilon \tilde{B})$$

denote the first-order condition in the local coordinates. This map is continuously differen-

tiable, satisfies $\Psi(\mathbf{0}; \mathbf{0}, 0) = \mathbf{0}$, and has

$$D_{\zeta}\Psi(\mathbf{0}; \mathbf{0}, 0) = -2 \operatorname{diag}(h_2, \dots, h_n),$$

which is nonsingular. The implicit-function theorem therefore yields a unique continuously differentiable branch $\zeta^*(\xi, \varepsilon)$ near $(\mathbf{0}, 0)$, with $\zeta^*(\mathbf{0}, 0) = \mathbf{0}$. The next two calculations give its partial derivatives at the benchmark; the joint expansion follows from its first-order Taylor approximation.

We now differentiate the local optimum with respect to initial disagreement, holding $\varepsilon = 0$. The terms in the local loss that depend on ζ are

$$\sum_{j>1} (h_j \zeta_j^2 + 2q_j \zeta_j \xi_j) + o(\|\zeta\|^2 + \|\zeta\| \|\xi\|).$$

The first-order condition in direction j is therefore

$$h_j \zeta_j + q_j \xi_j + o(\|\zeta\| + \|\xi\|) = 0.$$

Since $h_j > 0$,

$$\frac{\partial \zeta_j^*}{\partial \xi_j}(\mathbf{0}, 0) = -\frac{q_j}{h_j} \equiv M(\lambda_j),$$

while

$$\frac{\partial \zeta_j^*}{\partial \xi_l}(\mathbf{0}, 0) = 0 \quad \text{for } l \neq j.$$

Let $\tilde{k}(\lambda) = 1/(1 - \delta\lambda^2)$. Substituting the expressions for q_j and h_j gives

$$M(\lambda) = \frac{\lambda \tilde{k}(\lambda)(1 - \delta\tau^2)}{\tau \tilde{k}_1(1 - \delta\tau\lambda) + \ell_1 \tilde{k}(\lambda)(1 + \delta\tau\lambda)}.$$

The average-coordinate Riccati equation gives

$$\tilde{k}_1 = \frac{1}{1 - \delta\tau}, \quad \ell_1 = 1 - \tau.$$

The denominator in the expression for $M(\lambda)$ is positive on $[-1, 1]$, so $M(\lambda)$ has the sign of

λ . Multiplying numerator and denominator by $1 - \delta\lambda^2$, write

$$M(\lambda) = \frac{\lambda(1 - \delta\tau^2)}{N(\lambda)}, \quad N(\lambda) = \tau\tilde{k}_1(1 - \delta\tau\lambda)(1 - \delta\lambda^2) + (1 - \tau)(1 + \delta\tau\lambda).$$

Then

$$(1 - \delta\tau)(N(\lambda) - \lambda N'(\lambda)) = 1 - \delta\tau + \delta\tau^2 + \delta\tau\lambda^2 - 2\delta^2\tau^2\lambda^3 > 0.$$

For $\lambda \leq 0$ positivity is immediate. On $[0, 1]$, its derivative is $2\delta\tau\lambda(1 - 3\delta\tau\lambda)$, so any interior critical point is a maximum; the expression is strictly positive at both endpoints. Hence $M'(\lambda) > 0$ on $[-1, 1]$. Moreover, direct simplification gives, for $\lambda \in [0, 1]$,

$$\lambda - M(\lambda) = \frac{\delta\lambda\tau(1 - \lambda) [\lambda + 2\tau - \delta\tau(\lambda^2 + \lambda + \tau)]}{(1 - \delta\lambda^2)(1 - \delta\tau)\mathcal{D}(\lambda)},$$

where $\mathcal{D}(\lambda) > 0$ is the denominator in the expression for $M(\lambda)$. The term in brackets is concave in λ and strictly positive at both endpoints. Hence $M(0) = 0$, $M(1) = 1$ and $0 < M(\lambda) < \lambda$ for $\lambda \in (0, 1)$.

We next differentiate with respect to the network perturbation, holding $\boldsymbol{\xi} = \mathbf{0}$. Let

$$\tilde{B} = P'BP, \quad \tilde{\mathbf{s}} = \tilde{B}'\mathbf{e}_1 = P'\mathbf{s}, \quad \mathbf{s} = B'\mathbf{u}.$$

Because $B\mathbf{1} = \mathbf{0}$, $\tilde{B}\mathbf{e}_1 = \mathbf{0}$ and $\mathbf{e}_1'\tilde{\mathbf{s}} = 0$. Holding $\tilde{\mathbf{b}} = \mathbf{e}_1$, define

$$\dot{\tilde{K}} = \left. \frac{\partial \tilde{K}^*(\mathbf{e}_1; D + \varepsilon \tilde{B})}{\partial \varepsilon} \right|_{\varepsilon=0}$$

and let

$$\tilde{\mathbf{k}} = (I - \mathbf{e}_1\mathbf{e}_1')\dot{\tilde{K}}\mathbf{e}_1.$$

Differentiating the transformed Riccati equation at the benchmark and projecting its first column onto $\mathbf{e}_1^\perp$ gives

$$\tilde{\mathbf{k}} = \delta\tau\tilde{k}_1\tilde{B}'\mathbf{e}_1 + \delta\tau D\tilde{\mathbf{k}},$$

which gives $\tilde{\mathbf{k}} = \delta\tau\tilde{k}_1(I - \delta\tau D)^{-1}\tilde{B}'\mathbf{e}_1$. Define

$$\mathbf{g} = (I - \delta\tau A'_0)^{-1}\mathbf{s} = \sum_{j>1} g_j \mathbf{p}_j.$$

Because $\mathbf{e}_1' \tilde{\mathbf{s}} = 0$ and D preserves $\mathbf{e}_1^\perp$, we also have $g_1 = 0$. Since

$$g_j = \frac{\tilde{s}_j}{1 - \delta\tau\lambda_j}, \quad \tilde{s}_j = \mathbf{e}_j' \tilde{B}' \mathbf{e}_1,$$

we obtain, for every $j > 1$,

$$\dot{\tilde{K}}_{j1} = \delta\tau\tilde{k}_1 g_j.$$

We now differentiate the targeting condition. In the eigenbasis, the value gradient with respect to the exposure rule is

$$\nabla_{\tilde{\mathbf{b}}} \tilde{v} = 2\delta\tilde{K}\tilde{\tilde{A}}\tilde{\tilde{Z}}\tilde{\mathbf{p}}.$$

At the benchmark, this vector lies in $\text{span}\{\mathbf{e}_1\}$. Holding $\tilde{\mathbf{b}} = \mathbf{e}_1$, differentiation with respect to ε gives, for each $j > 1$,

$$\left. \frac{\partial}{\partial \varepsilon} \mathbf{e}_j' \nabla_{\tilde{\mathbf{b}}} \tilde{v} \right|_{(\boldsymbol{\xi}, \varepsilon) = (\mathbf{0}, 0)} = 2\delta\tau z_1 \ell_1 \dot{\tilde{K}}_{j1}.$$

Since $\tilde{B}\mathbf{e}_1 = \mathbf{0}$, the remaining benchmark factors map $\mathbf{e}_1$ into $\text{span}\{\mathbf{e}_1\}$. Hence the terms involving the derivatives of $\tilde{\tilde{A}}$, $\tilde{\tilde{Z}}$, and $\tilde{\mathbf{p}}$ remain in $\text{span}\{\mathbf{e}_1\}$ and vanish after projection onto $\mathbf{e}_j$. The change in the exposure rule contributes through the tangent Hessian. Differentiating the projected first-order condition along the local optimum therefore gives

$$-2h_j \frac{\partial \zeta_j^*}{\partial \varepsilon}(\mathbf{0}, 0) + 2\delta\tau z_1 \ell_1 \dot{\tilde{K}}_{j1} = 0.$$

Using $\dot{\tilde{K}}_{j1} = \delta\tau\tilde{k}_1 g_j$ and $z_1 = 1/(1 - \delta\tau^2)$, we obtain

$$\frac{\partial \zeta_j^*}{\partial \varepsilon}(\mathbf{0}, 0) = \chi(\lambda_j) g_j, \quad \text{where} \quad \chi(\lambda_j) = \frac{\delta^2 \tau^2 \ell_1 \tilde{k}_1}{(1 - \delta\tau^2) h_j} > 0.$$

Taking the total differential of the jointly differentiable branch gives, for each $j > 1$,

$$\zeta_j^*(\boldsymbol{\xi}, \varepsilon) = M(\lambda_j) \xi_j + \varepsilon \chi(\lambda_j) g_j + o(\|\boldsymbol{\xi}\| + |\varepsilon|).$$

Since $\tilde{\mathbf{b}}(\boldsymbol{\zeta}) = \mathbf{e}_1 + \boldsymbol{\zeta} + O(\|\boldsymbol{\zeta}\|^2)$, it follows that

$$\tilde{\mathbf{b}}^*(\boldsymbol{\xi}, \varepsilon) = \mathbf{e}_1 + \sum_{j>1} [M(\lambda_j) \xi_j + \varepsilon \chi(\lambda_j) g_j] \mathbf{e}_j + o(\|\boldsymbol{\xi}\| + |\varepsilon|).$$

Thus persistence and leverage add to first order; their interaction enters only at higher order.

Finally, we verify the clique claim following Proposition 4. Let $A_0 = J/n$ and consider the perturbation

$$B = \mathbf{1} \left(\mathbf{a} - \frac{1}{n} \mathbf{1} \right)'.$$

Then

$$\mathbf{s} = B' \mathbf{u} = \sqrt{n} \left(\mathbf{a} - \frac{1}{n} \mathbf{1} \right).$$

Because $\mathbf{s} \perp \mathbf{1}$ and $A_0' \mathbf{s} = \mathbf{0}$,

$$\mathbf{g} = (I - \delta \tau A_0')^{-1} \mathbf{s} = \mathbf{s}.$$

Moreover, every disagreement eigenvalue of A_0 is zero, so $\chi(\lambda_j) = \chi(0) > 0$ for every $j > 1$. Setting $\boldsymbol{\xi} = \mathbf{0}$ and returning to the original coordinates therefore gives

$$\mathbf{b}^*(\varepsilon) = \mathbf{u} + \varepsilon \chi \sqrt{n} \left(\mathbf{a} - \frac{1}{n} \mathbf{1} \right) + o(|\varepsilon|).$$

Equivalently, for some $C > 0$,

$$\mathbf{b}^*(\varepsilon) = \mathbf{u} + \varepsilon C \left(\mathbf{a} - \frac{1}{n} \mathbf{1} \right) + o(|\varepsilon|).$$

Hence $b_i^*(\varepsilon) - b_j^*(\varepsilon) = \varepsilon C(a_i - a_j) + o(|\varepsilon|)$, which recovers locally the pairwise authority pattern in Proposition 3. $\square$

A.3 Proofs from Section 5

Lemma 5. *Let $\mathbf{b} \in S^{n-1}$ be any coarse exposure rule. The optimality gap from coarse targeting under $\mathbf{b}$ to full targeting can be expressed as*

$$G(\mathbf{b}) = \sum_{t=0}^{\infty} \delta^t \tilde{A}'^t \Phi(\mathbf{b}) \tilde{A}^t, \quad \Delta(\mathbf{b}) = \mathbf{x}_0' G(\mathbf{b}) \mathbf{x}_0, \quad (10)$$

where $\Phi(\mathbf{b}) \equiv (\mathbf{b} \mathbf{p}' - F^f)'(cI + \delta K^f)(\mathbf{b} \mathbf{p}' - F^f) \geq 0$ is the gap source matrix.

Proof of Proposition 5: By Lemma 5, $G(\mathbf{b}) = \sum_{t=0}^{\infty} \delta^t \tilde{A}'^t \Phi(\mathbf{b}) \tilde{A}^t$ and $\Delta(\mathbf{b}) = \mathbf{x}_0' G(\mathbf{b}) \mathbf{x}_0$.

Therefore,

$$\Delta(\mathbf{b}) = \sum_{t=0}^{\infty} \delta^t \mathbf{x}'_0 \tilde{A}^{t'} \Phi(\mathbf{b}) \tilde{A}^t \mathbf{x}_0 = \sum_{t=0}^{\infty} \delta^t \text{tr} \left(\Phi(\mathbf{b}) \tilde{A}^t \mathbf{x}_0 \mathbf{x}'_0 \tilde{A}^{t'} \right) = \text{tr} \left(\Phi(\mathbf{b}) \sum_{t=0}^{\infty} \delta^t \tilde{A}^t \mathbf{x}_0 \mathbf{x}'_0 \tilde{A}^{t'} \right).$$

By definition, $Z(\mathbf{b}) = \sum_{t=0}^{\infty} \delta^t \tilde{A}^t \mathbf{x}_0 \mathbf{x}'_0 \tilde{A}^{t'}$. Thus $\Delta(\mathbf{b}) = \text{tr}(\Phi(\mathbf{b})Z(\mathbf{b}))$. $\square$

Proof of Proposition 6: Because the LQ problem is homogeneous of degree two in the initial state, $\Delta^*(s\mathbf{d}) = s^2\Delta^*(\mathbf{d})$. It therefore suffices to prove the result for $\|\mathbf{d}\|=1$. For a nearby row-sum-one matrix A , write

$$A_{\perp} = \rho\Pi + E, \quad \ell = \alpha\mathbf{d} + \mathbf{h},$$

where $\alpha = \ell'\mathbf{d}$ and $\mathbf{h}'\mathbf{d} = 0$.

Fix $\mathcal{S} = \text{span}\{\mathbf{d}\}$, let $v^{\mathcal{S}}$ denote the optimal value when intervention is restricted to $\mathcal{S}$, and define

$$G(E, \alpha, \mathbf{h}) = v^f - v^{\mathcal{S}} \geq 0.$$

When $E = 0$ and $\alpha = 0$, $A\mathbf{d} = \rho\mathbf{d}$, $A'\mathbf{d} = \rho\mathbf{d}$. Hence $\mathcal{S}$ and $\mathcal{S}^{\perp}$ are invariant under A , and, since $Q = I$, the full-targeting LQ problem separates across these two subspaces. Starting from $\mathbf{d}$, full targeting therefore intervenes only in $\mathcal{S}$, so $G(0, 0, \mathbf{h}) = 0$ for every nearby $\mathbf{h}$ orthogonal to $\mathbf{d}$.

For differentiation, work in the ambient affine space of nearby matrices satisfying $A\mathbf{1} = \mathbf{1}$, without imposing nonnegativity. Full targeting weakly dominates intervention restricted to $\mathcal{S}$ throughout this space, so $G \geq 0$. Moreover, $\rho(\sqrt{\delta}A_0) = \sqrt{\delta} < 1$, and thus on a sufficiently small neighborhood of A_0 , the full-target and restricted value functions are twice continuously differentiable in A . Thus, for every nearby $\mathbf{h}$, $(E, \alpha) = (0, 0)$ is an interior minimum of $G(\cdot, \cdot, \mathbf{h})$, implying

$$D_E G(0, 0, \mathbf{h}) = 0, \quad D_{\alpha} G(0, 0, \mathbf{h}) = 0.$$

A second-order Taylor expansion, uniformly for $\mathbf{h}$ near its benchmark value, therefore gives

$$G(E, \alpha, \mathbf{h}) \leq C (\|E\|^2 + \alpha^2)$$

for some $C > 0$. Since $\mathcal{S} = \text{span}\{\mathbf{d}\}$ is a feasible one-dimensional exposure space, optimal coarse targeting weakly improves on this restriction. Hence

$$\Delta^* \leq G(E, \alpha, \mathbf{h}) \leq C (\|E\|^2 + \alpha^2).$$

Restoring the scale of $\mathbf{d}$ and substituting $E = A_\perp - \rho\Pi$ and $\alpha = \frac{\ell'\mathbf{d}}{\|\mathbf{d}\|}$ yields the claim. $\square$

A.4 Proofs from Section 7

Proof of Proposition 7: We first compare myopic and full targeting. Matrix $\tilde{K}^f$ is the unique diagonal pd matrix solution to

$$\tilde{K}^f = I + \delta c D (\delta \tilde{K}^f + cI)^{-1} \tilde{K}^f D. \quad (11)$$

Matrix $\tilde{K}^f$ captures the decision-maker's future discounted loss, net of the cost of the optimal policy. Let $\tilde{K}_{jj}^f$ be the j th diagonal entry, which strictly increases in $|\lambda_j|$. Let $\Sigma = \sigma^2 I$. Since the eigenbasis transformation is orthogonal, the shock covariance remains Σ in the transformed coordinates. The decision-maker's value consists of two parts:

$$\tilde{v}^f(\tilde{\mathbf{x}}_0) = -\tilde{\mathbf{x}}_0' \tilde{K}^f \tilde{\mathbf{x}}_0 - \sum_{t=0}^{\infty} \delta^{t+1} \mathbb{E} \left(\tilde{\nu}_t' \tilde{K}^f \tilde{\nu}_t \right) = -\tilde{\mathbf{x}}_0' \tilde{K}^f \tilde{\mathbf{x}}_0 - \frac{\delta}{1-\delta} \text{tr}(\tilde{K}^f \Sigma). \quad (12)$$

For myopic targeting, without noise, the value is

$$-\tilde{\mathbf{x}}_0' \left(I - \frac{\delta c^2}{(\delta + c)^2} D^2 \right)^{-1} \left(I + \frac{c \delta^2}{(\delta + c)^2} D^2 \right) \tilde{\mathbf{x}}_0 = -\tilde{\mathbf{x}}_0' \tilde{K}^m \tilde{\mathbf{x}}_0, \quad (13)$$

where $\tilde{K}^m$ is the product of two matrices above, the first captures how each $\tilde{\mathbf{x}}_t$ depends on $\tilde{\mathbf{x}}_0$ given the optimal policy, and the second is how each stage payoff depends on $\tilde{\mathbf{x}}_t$. Clearly, $\tilde{K}^m$ is diagonal with entry $\tilde{K}_{jj}^m$, which strictly increases in $|\lambda_j|$. Each eigen-dimension is an independent scalar linear-quadratic problem. Full targeting uses the unique dynamically optimal scalar feedback, whereas the myopic feedback differs from it whenever $\lambda_j \neq 0$. Strict convexity of the scalar Bellman objective therefore implies $\tilde{k}_{jj}^f < \tilde{k}_{jj}^m$ whenever $\lambda_j \neq 0$, with equality when $\lambda_j = 0$. Moreover, the scalar Riccati equation implies that $\tilde{k}_{jj}^f$ is

increasing in λ_j^2 . Since $|\lambda_j| \leq 1 = \lambda_1$, we have $\tilde{k}_{jj}^f \leq \tilde{k}_{11}^f$. With noise, we have

$$\tilde{v}^m(\tilde{\mathbf{x}}_0) = -\tilde{\mathbf{x}}_0' \tilde{K}^m \tilde{\mathbf{x}}_0 - \frac{\delta}{1-\delta} \text{tr}(\tilde{K}^m \Sigma).$$

Let

$$\alpha = \frac{\delta \sigma^2}{1-\delta}, \quad \text{and} \quad \eta_j = \frac{\tilde{k}_{jj}^m - \tilde{k}_{jj}^f}{\tilde{k}_{jj}^f} \geq 0$$

for each j . The inequality is strict whenever $\lambda_j \neq 0$. Since

$$|\tilde{v}^f| = \sum_{j=1}^n ((\tilde{x}_0^j)^2 + \alpha) \tilde{k}_{jj}^f \quad \text{and} \quad |\tilde{v}^m| = \sum_{j=1}^n ((\tilde{x}_0^j)^2 + \alpha) \tilde{k}_{jj}^f (1 + \eta_j),$$

we can write

$$\frac{\tilde{v}^f}{\tilde{v}^m} = \frac{1}{1 + \sum_{j=1}^n \pi_j \eta_j},$$

where

$$\pi_j = \frac{((\tilde{x}_0^j)^2 + \alpha) \tilde{k}_{jj}^f}{\sum_{l=1}^n ((\tilde{x}_0^l)^2 + \alpha) \tilde{k}_{ll}^f}, \quad \sum_{j=1}^n \pi_j = 1.$$

Because A is row stochastic, $\lambda_1 = 1$, so $\eta_1 > 0$. Moreover, $\sigma > 0$ implies $\pi_1 > 0$. Hence $\sum_j \pi_j \eta_j > 0$, and therefore

$$0 < \frac{\tilde{v}^f}{\tilde{v}^m} = \frac{1}{1 + \sum_j \pi_j \eta_j} < 1.$$

Since both values are negative, $\tilde{v}^f > \tilde{v}^m$. Thus full targeting strictly dominates myopic targeting.

Second, we compare full and coarse targeting. Suppose first that $\text{rank}(A) = 1$. Since A is symmetric and row stochastic, $A = J/n$. Hence, in the eigenbasis of A , $D = \text{diag}(1, 0, \dots, 0)$. Under full targeting, the optimal feedback matrix is diagonal and has zero entries in every disagreement dimension. Thus full targeting intervenes only in the average-opinion dimension. The uniform coarse exposure rule $\tilde{\mathbf{b}} = \mathbf{e}_1$ implements exactly the same intervention. In the first dimension, the coarse and full Riccati equations coincide, while for every $j > 1$, $\tilde{k}_{jj}^c = \tilde{k}_{jj}^f = 1$. Therefore $\tilde{K}^c = \tilde{K}^f$ and $\tilde{v}^c(\tilde{\mathbf{x}}_0) = \tilde{v}^f(\tilde{\mathbf{x}}_0)$. Now suppose $\text{rank}(A) > 1$. Full targeting weakly dominates coarse targeting because every

coarse intervention $\tilde{\mathbf{b}}r_t$ is feasible under full targeting at the same cost. Under full targeting,

$$\tilde{\mathbf{u}}_t^f = -\tilde{F}^f \tilde{\mathbf{x}}_t, \quad \tilde{F}^f = \delta(cI + \delta\tilde{K}^f)^{-1}\tilde{K}^f D.$$

Since $\tilde{K}^f$ is pd and diagonal, $\text{rank}(\tilde{F}^f) = \text{rank}(D) = \text{rank}(A) > 1$. Thus the full-targeting intervention does not always lie in any fixed one-dimensional subspace. By contrast, under any coarse exposure rule $\tilde{\mathbf{b}}$, every intervention lies in $\text{span}\{\tilde{\mathbf{b}}\}$. For a fixed $\tilde{\mathbf{b}}$, coarse targeting can therefore match the full-targeting intervention only at states satisfying $\tilde{F}^f \tilde{\mathbf{x}}_t \in \text{span}\{\tilde{\mathbf{b}}\}$. Because $\text{rank}(\tilde{F}^f) > 1$, this set is a proper subspace of $\mathbb{R}^n$. Since $\sigma > 0$, the state has full support, so it lies outside this subspace with positive probability. At those states, no scalar r_t can reproduce the full-targeting intervention. Since the Bellman objective is strictly concave in the intervention vector, coarse targeting then yields a strictly lower payoff. Thus full targeting strictly dominates coarse targeting whenever $\text{rank}(A) > 1$. $\square$

Proof of Proposition 8: We show each of the two claims in turn.

Step 1: An optimal coarse exposure rule can be chosen in $\text{span}\{\mathbf{u}, \mathbf{d}\}$. Choose the eigenbasis so that its second vector is aligned with $\mathbf{d}$. Let

$$d = \|\mathbf{d}\|, \quad P = \left(\mathbf{u}, \frac{\mathbf{d}}{d}, \mathbf{p}_3, \dots, \mathbf{p}_n \right).$$

Then $\tilde{\mathbf{x}}_0 = P' \mathbf{x}_0 = (\sqrt{n} \bar{x}_0, d, 0, \dots, 0)'$ and $D = \text{diag}(1, \rho, \dots, \rho)$. Take any optimal exposure rule and write $\tilde{\mathbf{b}} = a\mathbf{e}_1 + b\mathbf{g}$, $\mathbf{g} \perp \mathbf{e}_1$ and $\|\mathbf{g}\| = 1$, where $b \geq 0$. If $b = 0$, the claim is true. Fixing a, b , the cost matrix has the block form

$$\tilde{K}^c(\mathbf{g}) = \begin{pmatrix} \bar{K}^c(a, b) & 0 \\ 0 & \tilde{k}_\rho^\emptyset I_{n-2} \end{pmatrix}$$

relative to the decomposition into $\text{span}\{\mathbf{e}_1, \mathbf{g}\}$ and its orthogonal complement. Because all disagreement directions have the same eigenvalue and shocks are isotropic, $\bar{K}^c(a, b)$ and $\text{tr} \tilde{K}^c(\mathbf{g})$ do not depend on the particular choice of $\mathbf{g}$.

Let $s = de'_2 \mathbf{g}$. The initial disagreement orthogonal to $\mathbf{g}$ has squared norm $d^2 - s^2$, so, up to terms independent of $\mathbf{g}$,

$$\tilde{v}(s) = -2\bar{K}_{12}^c(a, b)\sqrt{n} \bar{x}_0 s - \left(\bar{K}_{22}^c(a, b) - \tilde{k}_\rho^\emptyset \right) s^2.$$

Because zero intervention is feasible, $\bar{K}_{22}^c(a, b) \leq \tilde{k}_\rho^\emptyset$. Hence $\tilde{v}^c(s)$ is convex on $[-d, d]$ and is maximized at $s = d$ or $s = -d$. Thus an optimal disagreement direction can be chosen as $\mathbf{g} = \mathbf{e}_2$ or $\mathbf{g} = -\mathbf{e}_2$. Therefore an optimal exposure rule can be chosen in $\text{span}\{\mathbf{e}_1, \mathbf{e}_2\}$, or equivalently in $\text{span}\{\mathbf{u}, \mathbf{d}\}$.

Step 2: Asymptotic ranking. Consider a sequence of economies as $n \rightarrow \infty$. In the basis constructed in Step 1, $\tilde{\mathbf{x}}_0^{(n)} = (\mathbf{u}'_n \mathbf{x}_0^{(n)}, \|\mathbf{d}_n\|, 0, \dots, 0)'$. By assumption, $\|\tilde{\mathbf{x}}_0^{(n)}\| = O(1)$. For every disagreement dimension $j > 1$, the scalar cost coefficients under full targeting, myopic targeting, and no intervention are respectively $\tilde{k}_\rho^f$, $\tilde{k}_\rho^m$, and $\tilde{k}_\rho^\emptyset$. Then

$$\tilde{k}_\rho^f = 1 + \delta c \rho^2 \frac{\tilde{k}_\rho^f}{c + \delta \tilde{k}_\rho^f}, \quad \tilde{k}_\rho^m = \frac{1 + \frac{c\delta^2 \rho^2}{(c+\delta)^2}}{1 - \frac{\delta c^2 \rho^2}{(c+\delta)^2}}, \quad \tilde{k}_\rho^\emptyset = \frac{1}{1 - \delta \rho^2}.$$

Since $\rho \neq 0$, $0 < \tilde{k}_\rho^f < \tilde{k}_\rho^m < \tilde{k}_\rho^\emptyset$, because full targeting strictly improves on myopic targeting, and myopic targeting strictly improves on no intervention. Under full targeting,

$$-\tilde{v}_n^f = \tilde{k}_1^f (\mathbf{u}'_n \mathbf{x}_0^{(n)})^2 + \tilde{k}_\rho^f \|\mathbf{d}_n\|^2 + \frac{\delta \sigma^2}{1 - \delta} (\tilde{k}_1^f + (n-1)\tilde{k}_\rho^f).$$

Similarly, under myopic targeting,

$$-\tilde{v}_n^m = \tilde{k}_1^m (\mathbf{u}'_n \mathbf{x}_0^{(n)})^2 + \tilde{k}_\rho^m \|\mathbf{d}_n\|^2 + \frac{\delta \sigma^2}{1 - \delta} (\tilde{k}_1^m + (n-1)\tilde{k}_\rho^m).$$

The initial-state terms remain bounded by assumption. For coarse targeting, since the optimal exposure rule has support only on the first two coordinates,

$$\tilde{K}_n^c = \begin{pmatrix} \bar{K}_n^c & 0 \\ 0 & \tilde{k}_\rho^\emptyset I_{n-2} \end{pmatrix},$$

where $\bar{K}_n^c$ is the leading 2×2 block. Then

$$-\tilde{v}_n^c = \begin{pmatrix} \mathbf{u}'_n \mathbf{x}_0^{(n)} \\ \|\mathbf{d}_n\| \end{pmatrix}' \bar{K}_n^c \begin{pmatrix} \mathbf{u}'_n \mathbf{x}_0^{(n)} \\ \|\mathbf{d}_n\| \end{pmatrix} + \frac{\delta \sigma^2}{1 - \delta} (\text{tr } \bar{K}_n^c + (n-2)\tilde{k}_\rho^\emptyset).$$

Because zero intervention is feasible,

$$0 \prec \bar{K}_n^c \preceq \begin{pmatrix} \tilde{k}_1^\emptyset & 0 \\ 0 & \tilde{k}_\rho^\emptyset \end{pmatrix}.$$

Hence $\bar{K}_n^c$ is uniformly bounded in n . Since $\mathbf{u}'_n \mathbf{x}_0^{(n)} = O(1)$ and $\|\mathbf{d}_n\| = O(1)$,

$$-\tilde{v}_n^c = O(1) + \frac{\delta\sigma^2}{1-\delta} \left(\text{tr } \bar{K}_n^c + (n-2)\tilde{k}_\rho^\emptyset \right).$$

Therefore, dividing each value by n ,

$$\lim_{n \rightarrow \infty} \frac{\tilde{v}_n^f}{\tilde{v}_n^m} = \frac{\tilde{k}_\rho^f}{\tilde{k}_\rho^m}, \quad \lim_{n \rightarrow \infty} \frac{\tilde{v}_n^f}{\tilde{v}_n^c} = \frac{\tilde{k}_\rho^f}{\tilde{k}_\rho^\emptyset}.$$

Since $0 < \tilde{k}_\rho^f < \tilde{k}_\rho^m < \tilde{k}_\rho^\emptyset$, the conclusion follows.

The same calculation gives the fixed-subspace comparison stated in the text. Let $\tilde{v}_n^{(k_n)}$ denote the best value attainable with a fixed subspace of dimension k_n . Since the $n-1$ disagreement dimensions are symmetric, an optimal subspace controls $k_n + O(1)$ of them, where the $O(1)$ term allows for the average-opinion dimension. Initial-state terms remain $O(1)$, while the gain from controlling each disagreement dimension under recurrent shocks is the same. Hence

$$\frac{\tilde{v}_n^{(k_n)} - \tilde{v}_n^\emptyset}{\tilde{v}_n^f - \tilde{v}_n^\emptyset} = \frac{k_n + O(1)}{n-1} + o(1) \longrightarrow \alpha$$

whenever $k_n/n \rightarrow \alpha$. □

Supplementary Appendix

S1 Additional proofs

Proof of Lemma 1: Fix $\mathbf{b} \neq \mathbf{0}$, $c > 0$, and $\delta \in (0, 1)$. We verify that the discounted linear-quadratic problem has a unique stabilizing quadratic value function. First, the no-intervention policy is feasible and yields a finite value. Under $r_t \equiv 0$,

$$\sum_{t=0}^{\infty} \delta^t \mathbf{x}'_t Q \mathbf{x}_t = \mathbf{x}' \left(\sum_{t=0}^{\infty} (\sqrt{\delta} A')^t Q (\sqrt{\delta} A)^t \right) \mathbf{x}.$$

Since A is row-stochastic, $\rho(A) \leq 1$, so $\rho(\sqrt{\delta} A) < 1$. The series therefore converges, and the value is finite for all initial opinions. To apply the standard discrete-time LQ theorem, write the problem in scaled variables

$$\mathbf{y}_t = (\sqrt{\delta})^t \mathbf{x}_t, \quad u_t = (\sqrt{\delta})^t r_t, \quad \text{then}$$

$$\mathbf{y}_{t+1} = \sqrt{\delta} A \mathbf{y}_t + \sqrt{\delta} \mathbf{b} u_t, \text{ and } \sum_{t=0}^{\infty} \delta^t (\mathbf{x}'_t Q \mathbf{x}_t + c r_t^2) = \sum_{t=0}^{\infty} (\mathbf{y}'_t Q \mathbf{y}_t + c u_t^2).$$

Because $\sqrt{\delta} A$ is Schur stable, the pair $(\sqrt{\delta} A, \sqrt{\delta} \mathbf{b})$ is stabilizable. Since $Q \succ 0$, the pair $(Q^{1/2}, \sqrt{\delta} A)$ is detectable. With $c > 0$, the standard discrete-time LQ theorem gives a unique stabilizing quadratic value function $v(\mathbf{x}) = -\mathbf{x}' K^* \mathbf{x}$, where $K^* \succ 0$, and a unique linear optimal policy. It remains only to record the associated equations. The Bellman equation in the original discounted variables is

$$v(\mathbf{x}) = \max_r \left\{ -\mathbf{x}' Q \mathbf{x} - c r^2 - \delta (\mathbf{A} \mathbf{x} + \mathbf{b} r)' K^* (\mathbf{A} \mathbf{x} + \mathbf{b} r) \right\}.$$

The first-order condition gives

$$r^*(\mathbf{x}) = -\delta (c + \delta \mathbf{b}' K^* \mathbf{b})^{-1} \mathbf{b}' K^* \mathbf{A} \mathbf{x} \equiv -\mathbf{p}' \mathbf{x}.$$

Substituting this rule into the Bellman equation and matching quadratic terms gives

$$K^* = Q + \delta A' \left(K^* - \delta K^* \mathbf{b} (c + \delta \mathbf{b}' K^* \mathbf{b})^{-1} \mathbf{b}' K^* \right) A.$$

The stabilizing solution is unique and satisfies $\rho(\sqrt{\delta} \tilde{A}(\mathbf{b})) < 1$, so the Riccati equation uniquely determines the continuation value and the optimal intervention rule. $\square$

Proof of Lemma 2: Fix $\mathbf{b} \neq \mathbf{0}$ and let $\gamma > 1$. The dynamics under exposure rule $\gamma\mathbf{b}$ and intervention level $\tilde{r}_t$ are identical to those under exposure rule $\mathbf{b}$ and intervention level $r_t = \gamma\tilde{r}_t$. Hence replacing $\mathbf{b}$ by $\gamma\mathbf{b}$ is equivalent to lowering the intervention cost from c to c/γ^2 . The value weakly increases, and strictly increases whenever the optimal intervention is nontrivial. It remains to show that intervention is strictly valuable. Let K^0 denote the no-intervention loss matrix,

$$K^0 = \sum_{t=0}^{\infty} \delta^t (A')^t Q A^t.$$

Since $Q A \mathbf{x}_0 \neq \mathbf{0}$, we have $A \mathbf{x}_0 \neq \mathbf{0}$, and therefore $K^0 A \mathbf{x}_0 \neq \mathbf{0}$. Choosing an exposure direction proportional to $-K^0 A \mathbf{x}_0$ and a sufficiently small positive intervention at date zero, followed by zero intervention thereafter, strictly improves on no intervention. Hence every optimal exposure rule induces a nontrivial intervention. Therefore, if $\|\mathbf{b}\| < 1$, scaling $\mathbf{b}$ up slightly while remaining feasible strictly improves the value. Every optimal exposure rule must satisfy $\|\mathbf{b}^*\| = 1$. $\square$

To conduct local comparative statics, we call a stationary point *nondegenerate* if the Hessian of the Lagrangian restricted to the tangent space of the unit sphere is nonsingular.

Corollary S1. *Suppose every stationary point of the outer problem (5) is nondegenerate. Then stationary points are isolated and vary smoothly with the primitives; there are finitely many of them, and every local maximum is strict.*

Proof of Corollary S1: Fix a value of the scalar primitive θ and a stationary pair $(\mathbf{b}^*, \mu^*)$. Let $T_{\mathbf{b}^*} = \{\mathbf{h} \in \mathbb{R}^n : \mathbf{b}^{*'} \mathbf{h} = 0\}$ be the tangent space of the unit sphere at $\mathbf{b}^*$. Let $P^* \equiv I - \mathbf{b}^* \mathbf{b}^{*'}$ denote the orthogonal projection onto $T_{\mathbf{b}^*}$. Define the constrained Hessian $H(\mathbf{b}^*, \theta) : T_{\mathbf{b}^*} \rightarrow T_{\mathbf{b}^*}$ by

$$H(\mathbf{b}^*, \theta) \mathbf{h} = P^* (v_{\mathbf{b}\mathbf{b}}(\mathbf{b}^*; \theta) - \mu^* I) \mathbf{h}, \quad \mathbf{h} \in T_{\mathbf{b}^*}.$$

Equivalently,

$$H(\mathbf{b}^*, \theta) = P^* (v_{\mathbf{b}\mathbf{b}}(\mathbf{b}^*; \theta) - \mu^* I)|_{T_{\mathbf{b}^*}}.$$

By nondegeneracy, this operator is nonsingular on $T_{\mathbf{b}^*}$. The bordered derivative of the first-order condition and the constraint is

$$\begin{pmatrix} v_{\mathbf{b}\mathbf{b}}(\mathbf{b}^*; \theta) - \mu^* I & -\mathbf{b}^* \\ \mathbf{b}^{*'} & 0 \end{pmatrix}.$$

To see that this matrix is nonsingular, suppose $(\mathbf{h}, a)$ lies in its kernel. Then

$$\begin{aligned} (v_{\mathbf{b}\mathbf{b}}(\mathbf{b}^*; \theta) - \mu^* I) \mathbf{h} - a \mathbf{b}^* &= \mathbf{0}, \\ \mathbf{b}^{*'} \mathbf{h} &= 0. \end{aligned}$$

The second equation implies $\mathbf{h} \in T_{\mathbf{b}^*}$. Premultiplying the first equation by P^* gives $P^* (v_{\mathbf{b}\mathbf{b}}(\mathbf{b}^*; \theta) - \mu^* I) \mathbf{h} = \mathbf{0}$, because $P^* \mathbf{b}^* = \mathbf{0}$. By the definition of H , this is $H(\mathbf{b}^*, \theta) \mathbf{h} = \mathbf{0}$. Since $H(\mathbf{b}^*, \theta)$ is nonsingular on $T_{\mathbf{b}^*}$, $\mathbf{h} = \mathbf{0}$. The first equation then gives $a \mathbf{b}^* = \mathbf{0}$, and hence $a = 0$. The bordered derivative is therefore nonsingular. The implicit function theorem implies that each stationary exposure rule is locally unique and varies smoothly with the primitives. Nondegeneracy also implies that every stationary exposure rule is isolated. Since the stationary set is closed and contained in the compact unit sphere, it is finite. Finally, at a local maximum the constrained Hessian is negative semidefinite. Since it is nonsingular, it is negative definite, so every local maximum is strict. $\square$

Proof of Lemma 3: Fix the orthonormal eigenbasis used in the statement. Suppose by contradiction that an optimal exposure rule $\tilde{\mathbf{b}}^*$ has $\tilde{b}_j^* \neq 0$. For $\varepsilon \in (0, 1)$, define a nearby exposure rule $\hat{\mathbf{b}}$ by

$$\hat{b}_k = \tilde{b}_k^* \quad \text{for } k \neq j, \quad \hat{b}_j = (1 - \varepsilon) \tilde{b}_j^*.$$

Then $\|\hat{\mathbf{b}}\| < \|\tilde{\mathbf{b}}^*\| \leq 1$, so $\hat{\mathbf{b}}$ is feasible, distinct from $\tilde{\mathbf{b}}^*$, and arbitrarily close to it. Fix any intervention sequence $\{r_t\}$. All coordinates $k \neq j$ follow the same path under $\tilde{\mathbf{b}}^*$ and $\hat{\mathbf{b}}$. In coordinate j , the two dynamics are

$$\tilde{x}_{t+1}^j = \lambda_j \tilde{x}_t^j + \tilde{b}_j^* r_t, \quad \hat{x}_{t+1}^j = \lambda_j \hat{x}_t^j + (1 - \varepsilon) \tilde{b}_j^* r_t.$$

Since $\lambda_j \tilde{x}_0^j = 0$, we have $\hat{x}_1^j = (1 - \varepsilon) \tilde{x}_1^j$. Induction then gives

$$\hat{x}_t^j = (1 - \varepsilon) \tilde{x}_t^j \quad \text{for every } t \geq 1.$$

Thus loss in coordinate j is weakly lower at every date, and strictly lower whenever $\tilde{x}_t^j \neq 0$ for some $t \geq 1$. Losses in all other coordinates and intervention costs are unchanged.

Therefore every intervention sequence available under $\tilde{\mathbf{b}}^*$ yields a weakly higher payoff under $\hat{\mathbf{b}}$. Maximizing over intervention sequences gives

$$\tilde{v}(\tilde{\mathbf{x}}_0; \hat{\mathbf{b}}) \geq \tilde{v}(\tilde{\mathbf{x}}_0; \tilde{\mathbf{b}}^*).$$

If $\tilde{\mathbf{b}}^*$ is optimal, equality must hold, so $\hat{\mathbf{b}}$ is also optimal. But $\|\hat{\mathbf{b}}\| < 1$, contradicting Lemma 2, which implies that every optimal exposure rule has unit norm when $A\mathbf{x}_0 \neq \mathbf{0}$. Hence every optimal exposure rule satisfies $\tilde{b}_j^* = 0$. $\square$

Proof of Lemma 5: Define the gap matrix $G(\mathbf{b}) \equiv K^*(\mathbf{b}) - K^f$. We first show that $G(\mathbf{b})$ satisfies the Lyapunov equation

$$G(\mathbf{b}) = \Phi(\mathbf{b}) + \delta \tilde{A}' G(\mathbf{b}) \tilde{A},$$

and then establish a series representation.

Step 1: Riccati matrix identities. Recall that for any coarse exposure rule $\mathbf{b}$, we have $\mathbf{p} = \mathbf{p}(\mathbf{b})$ and $\tilde{A}(\mathbf{b}) = A - \mathbf{b}\mathbf{p}'$. Under the optimal scalar intervention level $r_t = -\mathbf{p}'\mathbf{x}_t$, the induced vector intervention is $\mathbf{u}_t = -\mathbf{b}\mathbf{p}'\mathbf{x}_t$, and the coarse transition matrix is $\tilde{A} = A - \mathbf{b}\mathbf{p}'$. The value iteration for the coarse problem can be rewritten as

$$K^*(\mathbf{b}) = Q + c\mathbf{p}\mathbf{p}' + \delta \tilde{A}' K^*(\mathbf{b}) \tilde{A}. \quad (\text{S1})$$

The term $c\mathbf{p}\mathbf{p}'$ comes from the intervention cost

$$cr_t^2 = c(\mathbf{p}'\mathbf{x}_t)^2 = \mathbf{x}_t' c\mathbf{p}\mathbf{p}' \mathbf{x}_t.$$

Under full targeting, the optimal vector intervention is $\mathbf{u}_t = -F^f \mathbf{x}_t$, where

$$F^f = \delta(cI + \delta K^f)^{-1} K^f A.$$

The full-targeting value iteration satisfies

$$K^f = Q + c(F^f)' F^f + \delta(A - F^f)' K^f (A - F^f).$$

We use the standard LQ completion-of-squares identity. For any feedback matrix $F \in \mathbb{R}^{n \times n}$,

$$cF'F + \delta(A - F)'K^f(A - F) = K^f - Q + (F - F^f)'(cI + \delta K^f)(F - F^f). \quad (\text{S2})$$

To verify this identity, expand the right-hand side and use

$$(cI + \delta K^f)F^f = \delta K^f A, \quad (F^f)'(cI + \delta K^f) = \delta A' K^f,$$

together with the Riccati equation for K^f . Apply (S2) to the coarse feedback matrix $F^c(\mathbf{b}) = \mathbf{b}\mathbf{p}'$. Since $\|\mathbf{b}\| = 1$, we have $cF'F = c(\mathbf{b}\mathbf{p}')'(\mathbf{b}\mathbf{p}') = c\mathbf{p}\mathbf{b}'\mathbf{b}\mathbf{p}' = c\mathbf{p}\mathbf{p}'$. Hence

$$c\mathbf{p}\mathbf{p}' + \delta\tilde{A}'K^f\tilde{A} = K^f - Q + \Phi(\mathbf{b}), \quad (\text{S3})$$

where $\Phi(\mathbf{b}) \equiv (F^c(\mathbf{b}) - F^f)'(cI + \delta K^f)(F^c(\mathbf{b}) - F^f)$. Since $cI + \delta K^f > 0$, we have $\Phi(\mathbf{b}) \geq 0$.

Step 2: gap Lyapunov representation. Now return to the coarse identity (S1). Add and subtract $\delta\tilde{A}'K^f\tilde{A}$:

$$K^*(\mathbf{b}) = Q + \left[c\mathbf{p}\mathbf{p}' + \delta\tilde{A}'K^f\tilde{A} \right] + \delta\tilde{A}'(K^*(\mathbf{b}) - K^f)\tilde{A}.$$

Using (S3) and canceling Q gives

$$K^*(\mathbf{b}) = K^f + \Phi(\mathbf{b}) + \delta\tilde{A}'(K^*(\mathbf{b}) - K^f)\tilde{A}. \quad (\text{S4})$$

Therefore, with $G(\mathbf{b}) = K^*(\mathbf{b}) - K^f$,

$$G(\mathbf{b}) = \Phi(\mathbf{b}) + \delta\tilde{A}'G(\mathbf{b})\tilde{A}. \quad (\text{S5})$$

Step 3: Series representation. It remains to iterate (S5). Repeated substitution gives, for every $T \geq 0$,

$$G(\mathbf{b}) = \sum_{t=0}^T \delta^t \tilde{A}^{t'} \Phi(\mathbf{b}) \tilde{A}^t + \delta^{T+1} \tilde{A}^{(T+1)'} G(\mathbf{b}) \tilde{A}^{T+1}.$$

Because $\rho(\sqrt{\delta} \tilde{A}) < 1$, the final term converges to zero as $T \rightarrow \infty$. Hence

$$G(\mathbf{b}) = \sum_{t=0}^{\infty} \delta^t \tilde{A}' \Phi(\mathbf{b}) \tilde{A}^t.$$

Finally, $\Delta(\mathbf{b}) = v^f(\mathbf{x}_0) - v(\mathbf{x}_0; \mathbf{b}) = \mathbf{x}_0'(K^*(\mathbf{b}) - K^f)\mathbf{x}_0 = \mathbf{x}_0'G(\mathbf{b})\mathbf{x}_0$ as claimed. $\square$

Proof of Lemma 4: Suppose first that the $\mathcal{S}$ -restricted technology attains the full-targeting value from $\mathbf{x}_0$. Since the optimal unrestricted intervention at each state is unique, the restricted policy must coincide with the full-targeting policy along the resulting path. Hence $F^f \mathbf{x}_t^f \in \mathcal{S}$ for every $t \geq 0$.

Conversely, suppose $F^f \mathbf{x}_t^f \in \mathcal{S}$ for every $t \geq 0$. At $t = 0$, the intervention $\mathbf{u}_0 = -F^f \mathbf{x}_0^f$ is feasible under the $\mathcal{S}$ -restricted technology and therefore reproduces $\mathbf{x}_1^f$. Inductively, if the restricted and full-targeting states agree at date t , then $\mathbf{u}_t = -F^f \mathbf{x}_t^f$ is feasible and reproduces $\mathbf{x}_{t+1}^f$. Thus the full-targeting intervention path is feasible under the $\mathcal{S}$ restriction, so the restricted technology attains $v^f(\mathbf{x}_0)$. $\square$

Proof of Example 1: Under full targeting with $Q = I$, the problem separates across the eigendirections of a symmetric network. For a scalar eigendirection with network eigenvalue $\lambda > 0$, let $m(\lambda)$ denote its optimal closed-loop persistence rate. If $k(\lambda)$ is the corresponding scalar Riccati coefficient, then

$$m(\lambda) = \frac{c\lambda}{c + \delta k(\lambda)},$$

and the scalar Riccati equation implies

$$c\delta\lambda m^2 - (c + \delta + c\delta\lambda^2)m + c\lambda = 0.$$

Write the left-hand side as $P(\lambda, m)$. At the optimal root, $0 < m(\lambda) < \lambda$: indeed,

$$P(\lambda, 0) = c\lambda > 0, \quad P(\lambda, \lambda) = -\delta\lambda < 0.$$

Hence this is the smaller root of the quadratic in m , so $P_m < 0$. Moreover,

$$P_\lambda = c(1 + \delta m^2 - 2\delta\lambda m) = c(1 - \delta\lambda^2 + \delta(m - \lambda)^2) > 0.$$

Implicit differentiation therefore gives $m'(\lambda) = -\frac{P_\lambda}{P_m} > 0$. Thus distinct positive network

eigenvalues induce distinct optimal closed-loop persistence rates.

At $A(\rho)$, with $0 < \rho < 1$, the only network eigenvalues are 1 and ρ . Starting from

$$\mathbf{x}_0 = (\mathbf{u}'\mathbf{x}_0)\mathbf{u} + \mathbf{d},$$

full targeting therefore remains in $\text{span}\{\mathbf{u}, \mathbf{d}\}$. Since both components are nonzero, $m(1) \neq m(\rho)$ and the intervention coefficient $\lambda - m(\lambda)$ is positive in both directions. Hence the full-targeting intervention path spans exactly $\text{span}\{\mathbf{u}, \mathbf{d}\}$, so $k_{\min} = 2$.

Now choose an orthonormal disagreement basis $\{\mathbf{q}_j\}_{j=2}^n$ with $\mathbf{q}_j'\mathbf{d} \neq 0$ for every j , choose distinct scalars γ_j , and let

$$A_\varepsilon = A(\rho) + \varepsilon \sum_{j=2}^n \gamma_j \mathbf{q}_j \mathbf{q}_j'.$$

For sufficiently small ε , A_ε remains symmetric, stochastic, and nonnegative, and all of its eigenvalues

$$1, \rho + \varepsilon\gamma_2, \dots, \rho + \varepsilon\gamma_n$$

remain positive. For nonzero sufficiently small ε , they are distinct, so the corresponding closed-loop persistence rates

$$m(1), m(\rho + \varepsilon\gamma_2), \dots, m(\rho + \varepsilon\gamma_n)$$

are also distinct.

Let $f(\lambda) = \lambda - m(\lambda) > 0$ denote the full-targeting feedback coefficient in an eigendirection. Since the initial state has a nonzero component in every eigendirection, the first n full-targeting interventions have coefficient matrix, up to nonzero diagonal factors,

$$\begin{pmatrix} 1 & m_1 & m_1^2 & \cdots & m_1^{n-1} \\ 1 & m_2 & m_2^2 & \cdots & m_2^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & m_n & m_n^2 & \cdots & m_n^{n-1} \end{pmatrix},$$

where the m_j are distinct. This Vandermonde matrix is nonsingular. The full-targeting

intervention path therefore spans $\mathbb{R}^n$, and

$$k_{\min} = n.$$

Because ε can be arbitrarily small, exact dimension jumps from 2 to n arbitrarily close to $A(\rho)$. $\square$

S2 Persistent bias and coarse targeting

Consider a time-invariant bias,¹⁵

$$\mathbf{x}_{t+1} = A\mathbf{x}_t + \mathbf{b}r_t + \mathbf{w}.$$

Unlike recurrent shocks, the same source is injected every period. Its effect is therefore predictable: social updating propagates it along the orbit

$$\mathbf{w}, A\mathbf{w}, A^2\mathbf{w}, \dots$$

For any fixed exposure rule, augmenting the state by the constant vector $\mathbf{w}$ turns the problem into a standard discounted linear-quadratic problem. Hence the value is quadratic in the augmented state and the optimal intervention is affine in current opinions.

Proposition S1. *Suppose A is symmetric and $Q = I$. Let $\Pi = I - \mathbf{u}\mathbf{u}'$, $\mathbf{d} = \Pi\mathbf{x}_0$, and $\mathbf{w}_\perp = \Pi\mathbf{w}$. Define*

$$\mathcal{H}(\mathbf{x}_0, \mathbf{w}) = \text{span} \{ \mathbf{u}, A^{t+1}\mathbf{d}, A^t\mathbf{w}_\perp : t \geq 0 \}.$$

Then every strict local optimum satisfies $\mathbf{b}^ \in \mathcal{H}(\mathbf{x}_0, \mathbf{w})$.*

The subspace contains average opinion, inherited disagreement that survives at least one update, and the predictable orbit of newly injected disagreement. Thus persistence of a source does not by itself make the targeting problem high-dimensional; what matters is the dimension of the orbit generated by that source.

For the symmetric benchmark $A(\rho)$, the implication is immediate.

¹⁵The vector $\mathbf{w}$ can represent persistent misinformation, agent-level bias, or anchored opinions.

Corollary S2. Suppose $Q = I$ and $A = A(\rho)$. Then every strict local optimum satisfies $\mathbf{b}^* \in \text{span}\{\mathbf{u}, \rho\mathbf{d}, \mathbf{w}_\perp\}$.

Under $A(\rho)$, $A^{t+1}\mathbf{d} = \rho^{t+1}\mathbf{d}$ and $A^t\mathbf{w}_\perp = \rho^t\mathbf{w}_\perp$, so Proposition S1 gives $\mathcal{H}(\mathbf{x}_0, \mathbf{w}) = \text{span}\{\mathbf{u}, \rho\mathbf{d}, \mathbf{w}_\perp\}$.

Proof of Proposition S1: Write $\mathcal{H} = \mathcal{H}(\mathbf{x}_0, \mathbf{w})$. Since A is symmetric, both $\mathcal{H}$ and $\mathcal{H}^\perp$ are invariant under A , and

$$A\mathbf{x}_0 + \mathbf{w} \in \mathcal{H}.$$

Thus, from date one onward, every component of opinions in $\mathcal{H}^\perp$ is generated only by intervention. Any strict local optimum must have unit norm. Otherwise, if intervention is nontrivial, scaling the exposure rule outward implements the same opinion path at lower intervention cost; if intervention is always zero, nearby exposure rules attain the same payoff.

Suppose therefore that a unit exposure rule has a nonzero component in $\mathcal{H}^\perp$. Write

$$\mathbf{b} = q\mathbf{h} + s\mathbf{k}, \quad q \geq 0, \quad s > 0, \quad q^2 + s^2 = 1,$$

where $\mathbf{h} \in \mathcal{H}$ and $\mathbf{k} \in \mathcal{H}^\perp$ are unit vectors. If $q = 0$, choose any unit $\mathbf{h} \in \mathcal{H}$. For $q' > q$ sufficiently close to q , let

$$s' = \sqrt{1 - q'^2}, \quad \mathbf{b}' = q'\mathbf{h} + s'\mathbf{k}, \quad r'_t = \frac{q}{q'}r_t.$$

The intervention inside $\mathcal{H}$ is unchanged,

$$q'\mathbf{h}r'_t = q\mathbf{h}r_t,$$

while the intervention in $\mathcal{H}^\perp$ is reduced:

$$s'\mathbf{k}r'_t = \frac{s'q}{sq'}s\mathbf{k}r_t, \quad 0 \leq \frac{s'q}{sq'} < 1.$$

Because $\mathcal{H}$ and $\mathcal{H}^\perp$ are invariant under A , the opinion path inside $\mathcal{H}$ is unchanged, while its $\mathcal{H}^\perp$ component is weakly reduced. Intervention cost is also weakly lower, with strict inequality whenever $r_t \neq 0$. Hence $\mathbf{b}'$ yields a strictly higher payoff whenever intervention

is nontrivial; if intervention is identically zero, it yields the same payoff. Either case contradicts strict local optimality. Therefore $\mathbf{b}^* \in \mathcal{H}$. $\square$

S3 Nonlinear updating and local disagreement dynamics

The effectiveness of coarse targeting depends on how disagreement propagates, rather than on the specific micro-foundation for opinion updating. For a smooth updating rule $\mathbf{x}_{t+1} = F(\mathbf{x}_t) + \mathbf{b}r_t$, consider local deviations around a no-intervention fixed point $\bar{\mathbf{x}}$, satisfying $F(\bar{\mathbf{x}}) = \bar{\mathbf{x}}$. Then

$$\Delta \mathbf{x}_{t+1} \approx J(\bar{\mathbf{x}}) \Delta \mathbf{x}_t + \mathbf{b}r_t,$$

where $J(\bar{\mathbf{x}})$ is the Jacobian of F evaluated at $\bar{\mathbf{x}}$. Thus the relevant local object is the propagation matrix $J(\bar{\mathbf{x}})$. The baseline model is the special case in which this matrix is constant and equal to A . As an illustration, consider the Friedkin–Johnsen model (Friedkin and Johnsen, 1990),

$$\mathbf{x}_{t+1} = (I - \Lambda)W\mathbf{x}_t + \Lambda\mathbf{x}_0 + \mathbf{b}r_t,$$

where W is row-stochastic and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$, with $\lambda_i \in [0, 1]$. Let $\bar{\mathbf{x}}$ be the steady state absent intervention:

$$\bar{\mathbf{x}} = (I - \Lambda)W\bar{\mathbf{x}} + \Lambda\mathbf{x}_0.$$

Defining deviations from this steady state by $\Delta \mathbf{x}_t = \mathbf{x}_t - \bar{\mathbf{x}}$, we obtain

$$\Delta \mathbf{x}_{t+1} = (I - \Lambda)W\Delta \mathbf{x}_t + \mathbf{b}r_t.$$

Hence the model takes the same local form as the baseline model, with $A = (I - \Lambda)W$. Here A is substochastic when some agents are partially stubborn. If stubbornness anchors every recurrent component of the network, then $\rho(A) < 1$, so deviations decay absent intervention. Components associated with eigenvalues close to one, however, decay slowly and therefore behave like persistent disagreement for a patient decision-maker. Since Friedkin–Johnsen updating is affine, this representation is exact rather than merely local.

The same local logic applies to nonlinear models with homophily (Golub and Jackson, 2012) or persuasion bias (DeMarzo et al., 2003). Around a fixed point, the Jacobian determines which disagreement dimensions propagate and how slowly they decay. If the local propagation matrix has the island structure, the relevant dimensions are group

disagreements as in the island model; if it has clustered structure, coarse targeting responds locally to slow-moving cluster disagreements. Thus the mechanism is robust within the region where the linear approximation is accurate: coarse targeting prioritizes disagreement dimensions that decay slowly, even when all disagreement is ultimately transient.

S4 Computational algorithm

S4.1 Comparison of algorithms

In general, the optimal coarse targeting weights do not admit a closed-form solution, so computation relies on numerical methods that exploit the fixed-point characterization in equation (7). This appendix describes the numerical methods used to search for an optimal exposure rule. We consider two approaches. The first is a *nested fixed-point* algorithm directly implied by the first-order condition in (7), which evaluates the gradient efficiently by solving the Lyapunov equation $(\sqrt{\delta}\tilde{A})Z(\sqrt{\delta}\tilde{A})' - Z + \mathbf{x}_0\mathbf{x}_0' = 0$. Related nested fixed-point algorithms are common in dynamic discrete choice problems (Rust, 1987). The second is a class of *projected gradient methods* (PGM) for constrained nonconvex optimization, which serve as a natural benchmark and are widely used in related settings (Gao et al., 2021; Solimine and Meyer-Baese, 2022).

Both approaches require repeated evaluation of the value function and its gradient. For a given $\mathbf{b}$, this involves solving an algebraic Riccati equation for the continuation value $K^*(\mathbf{b})$ and a discrete Lyapunov equation for the associated Gramian (Vaughan, 1970; Laub, 1979).

A. Common computational primitives. Both the nested fixed-point iteration and the projected gradient methods rely on the same underlying computational primitives. For any candidate exposure vector $\mathbf{b}$, the continuation value $K^*(\mathbf{b})$ is obtained as the unique stabilizing solution to an algebraic Riccati equation. Conditional on $K^*(\mathbf{b})$, the induced feedback vector $\mathbf{p}(\mathbf{b})$ and transition matrix $\tilde{A}(\mathbf{b}) = A - \mathbf{b}\mathbf{p}(\mathbf{b})'$ are computed, and the discounted Gramian associated with the induced opinion dynamics is obtained by solving a discrete Lyapunov equation. Together, these objects determine the gradient $\nabla_{\mathbf{b}}v(\mathbf{b})$ appearing on the right-hand side of the first-order condition (7).¹⁶

¹⁶The gradient formula is valid for any symmetric positive-definite payoff matrix Q . The derivation does not differentiate Q with respect to $\mathbf{b}$; Q enters through the Riccati solution $K^*(\mathbf{b})$. Thus the same algorithm applies to the population-weighted specification Q^{pop} .

Algorithm 3 summarizes the steps required to compute $\nabla_{\mathbf{b}}v(\mathbf{b})$ for a given $\mathbf{b}$. Both classes of algorithms discussed below repeatedly call this routine; they differ only in how the exposure vector $\mathbf{b}$ is updated across iterations, not in the evaluation of the underlying matrix equations.

B. Projected gradient method (benchmark). As a benchmark, we implement a projected gradient method (PGM) for the constrained maximization of the value function $v(\mathbf{b})$ subject to the norm constraint $\|\mathbf{b}\|=1$. Projected gradient methods are standard tools for constrained, nonconvex optimization problems and have been used in related targeting environments (Gao et al., 2021).

Starting from an initial exposure vector $\mathbf{b}_0$, the method updates $\mathbf{b}$ iteratively in the direction of the gradient $\nabla_{\mathbf{b}}v(\mathbf{b}_k)$, projected onto the tangent space of the constraint set. Concretely, each update takes the form

$$\mathbf{b}_{k+1} = \frac{\mathbf{b}_k + \eta_k(I - \mathbf{b}_k\mathbf{b}_k')\nabla_{\mathbf{b}}v(\mathbf{b}_k)}{\|\mathbf{b}_k + \eta_k(I - \mathbf{b}_k\mathbf{b}_k')\nabla_{\mathbf{b}}v(\mathbf{b}_k)\|},$$

where $(I - \mathbf{b}_k\mathbf{b}_k')$ removes the component of the gradient that points in the direction of $\mathbf{b}_k$, and η_k is a step size.

We use an adaptive step size: η_k is selected by a line search along the projected-gradient direction to maximize the value function locally. Fixed-step implementations are sensitive to the selected step size, so the reported comparison uses adaptive PGM. Algorithm 1 summarizes the procedure. Under standard regularity conditions, PGM converges to a stationary point of the constrained problem, though convergence is generally local and may depend on initialization.

C. Nested fixed-point iteration. The second approach is a nested fixed-point iteration. Rather than updating $\mathbf{b}$ along a gradient direction with an explicit step size, this method constructs a sequence $\{\mathbf{b}_k\}$ by repeatedly evaluating the fixed-point mapping implied by the optimality condition.

Starting from an initial exposure vector $\mathbf{b}_0$, each iteration updates $\mathbf{b}$ by plugging the current iterate into the right-hand side of the first-order condition and renormalizing to satisfy the norm constraint,

$$\mathbf{b}_{k+1} \propto \nabla_{\mathbf{b}}v(\mathbf{b}_k),$$

with proportionality chosen so that $\|\mathbf{b}_{k+1}\| = 1$. Algorithm 2 summarizes this update rule.

The method is *nested* because evaluation of the fixed-point mapping at $\mathbf{b}_k$ requires solving for the continuation value $K^*(\mathbf{b}_k)$ and the associated discounted Gramian, as described in Part A. In contrast to the projected gradient method, the nested fixed-point iteration does not require choosing a step size or performing a line search; the update is entirely determined by the fixed-point characterization of optimal targeting. For a given iterate $\mathbf{b}_k$, the matrix equation defining $K^*(\mathbf{b}_k)$ is solved directly using a standard Schur decomposition–based algorithm for algebraic Riccati equations (Laub, 1979). This avoids iterative solution of the inner fixed point and allows each outer iteration to be computed reliably using well-established numerical linear algebra routines.

Algorithm 1: Adaptive PGM

Data: A , $\mathbf{x}_0$, δ , c , Q

Result: Stationary candidate exposure vector $\hat{\mathbf{b}}$

$\mathbf{b}_0 \leftarrow \frac{A\mathbf{x}_0}{\|A\mathbf{x}_0\|}$ if $A\mathbf{x}_0 \neq \mathbf{0}$; otherwise
 $\mathbf{b}_0 \leftarrow \mathbf{e}_1$ $k \leftarrow 0$;

while not converged do

$\mathbf{g}_k \leftarrow \nabla_{\mathbf{b}} v(\mathbf{b}_k)$;
 $\mathbf{g}_k^\perp \leftarrow (I - \frac{\mathbf{b}_k \mathbf{b}_k'}{\mathbf{b}_k' \mathbf{b}_k}) \mathbf{g}_k$;
Define $\mathbf{f}_k(\eta) = \frac{\mathbf{b}_k + \eta \mathbf{g}_k^\perp}{\|\mathbf{b}_k + \eta \mathbf{g}_k^\perp\|}$;
 $\eta_k^* \leftarrow \arg \max_{\eta} v(\mathbf{f}_k(\eta))$;
 $\mathbf{b}_{k+1} \leftarrow \mathbf{f}_k(\eta_k^*)$;
 $k \leftarrow k + 1$;

end

return $\mathbf{b}_k$;

Algorithm 2: Nested fixed-point

Data: A , $\mathbf{x}_0$, δ , c , Q

Result: Stationary candidate exposure vector $\hat{\mathbf{b}}$

$\mathbf{b}_0 \leftarrow \frac{A\mathbf{x}_0}{\|A\mathbf{x}_0\|}$ if $A\mathbf{x}_0 \neq \mathbf{0}$; otherwise
 $\mathbf{b}_0 \leftarrow \mathbf{e}_1$ $k \leftarrow 0$;

while not converged do

$\mathbf{b}_{k+1} \leftarrow \nabla_{\mathbf{b}} v(\mathbf{b}_k)$;
 $\mathbf{b}_{k+1} \leftarrow \mathbf{b}_{k+1} / \|\mathbf{b}_{k+1}\|$;
 $k \leftarrow k + 1$;

end

return $\mathbf{b}_k$;

Algorithm 3: Compute gradient $\nabla_{\mathbf{b}} v(\mathbf{b})$

Data: $\mathbf{b}$, A , δ , c , Q

Result: Gradient $\nabla_{\mathbf{b}} v(\mathbf{b})$

Solve $K^* = Q + \delta A' K^* (I + \frac{\delta}{c} \mathbf{b} \mathbf{b}' K^*)^{-1} A$ for K^* ;

$\mathbf{p}' \leftarrow (c + \delta \mathbf{b}' K^* \mathbf{b})^{-1} \delta \mathbf{b}' K^* A$;

$\tilde{A} \leftarrow A - \mathbf{b} \mathbf{p}'$;

Solve $\sqrt{\delta} \tilde{A} Z \sqrt{\delta} \tilde{A}' - Z = -\mathbf{x}_0 \mathbf{x}_0'$ for Z ;

return $2\delta K^* \tilde{A} Z \mathbf{p}$;

D. Practical implementation details. Both algorithms require an initial exposure vector and a stopping criterion. Before comparing successive directions, we choose the sign so that $\mathbf{b}'_{k+1} \mathbf{b}_k \geq 0$. In the algorithm-comparison simulations, iterations terminate when the sign-aligned iterate distance satisfies $\|\mathbf{b}_{k+1} - \mathbf{b}_k\|_1 < 10^{-6}$ or after 200 iterations. In the empirical adaptive-PGM implementation, the projected-gradient tolerance is 10^{-9} , the line-search tolerance is 10^{-7} , the objective-improvement tolerance is 10^{-10} , and the cap is 60 outer iterations. The multi-start initializations used for the application are documented in Appendix S5.2. The fixed-point map is fast when stable: it converges in the opening island example, the $Q = I$ application, and most simulation instances. The population-weighted empirical application instead uses adaptive projected gradient, because the undamped fixed-point iteration frequently cycles from both informed and random starts when $Q = Q^{\text{pop}}$.

E. Simulation design. To evaluate the numerical performance of the algorithms described above, we conduct simulations on synthetic networks of varying size and structure. The goal of these experiments is not to assess economic outcomes, but to compare convergence behavior and computational cost across methods under controlled conditions.

We use two simulation panels. The algorithm-comparison panel uses $n \in \{2, 4, \dots, 32\}$ and 25 independent networks at each size. A separate scalability panel uses $n \in \{25, 50, 100, 150, 200, 300, 400, 600\}$ and 10 independent networks at each size. Adaptive PGM is run through $n = 200$, while the fixed-point method is run for every listed size. Each network is drawn from a uniform Dirichlet ensemble: each row of A is sampled independently from a Dirichlet distribution on the probability simplex, ensuring that A is row-stochastic and dense. This design produces asymmetric networks with heterogeneous influence patterns, similar in spirit to the empirical Facebook network used in the application.

For each network, we draw an initial opinion vector $\mathbf{x}_0$ as follows:

$$\mu_s \sim \mathcal{N}(0, 5), \quad \sigma_s \sim \text{Uniform}(0, 5), \quad \mathbf{x}_0 \sim \mathcal{N}(\mu_s \mathbf{1}, \sigma_s^2 I).$$

This construction generates both low- and high-dispersion initial conditions, allowing us to assess algorithm performance across a range of disagreement profiles.

F. Results. Figure 4 summarizes the comparison panel. It reports wall time and the number of iterations required for convergence as a function of network size, averaged across simulation draws. In the comparison panel, nested fixed-point averages 4.13 outer iterations

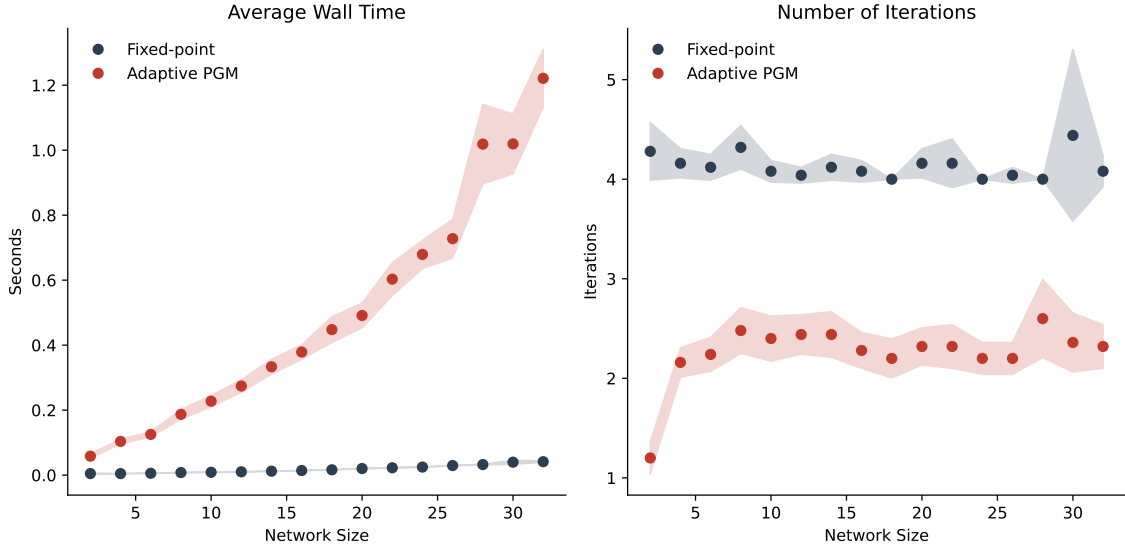


Figure 4: Convergence speed in the algorithm-comparison panel ($n \leq 32$)

and 0.018 seconds per completed instance (0.0045 seconds per iteration)¹⁷. Adaptive PGM averages fewer iterations (2.26) but 0.49 seconds per completed instance (0.217 seconds per iteration), due to the expensive line search operation. The scalability panel shows the same efficiency ranking. The fixed-point method averages approximately four iterations through $n = 600$ and 5.1 iterations at $n = 1000$. At $n = 200$, it converges in 34 seconds on average, compared with 1,032 seconds for the adaptive PGM. Each iteration of both methods involves solving matrix equations with $\mathcal{O}(n^3)$ complexity. Adaptive PGM additionally incurs repeated value-function evaluations during line search, leading to a per-iteration cost of

¹⁷Simulations were run on a 13th Gen Intel Core i7-1355U at 1700 Mhz.

order $\mathcal{O}(\log(1/\epsilon) n^3)$, where ϵ denotes the line-search tolerance.¹⁸ This difference reflects the fact that the fixed-point update directly implements the first-order condition, whereas gradient-based methods perform a sequence of local ascent and line-search steps.

When the fixed-point mapping is unstable, adaptive projected gradient provides the robust fallback. In simulation instances where both methods converge, their returned directions agree within the stopping tolerance. The comparison therefore highlights a trade-off between speed and robustness rather than differences in optimality.

S5 Data appendix

S5.1 Proof and construction details for Section 6.1

The model presented throughout the paper takes the opinion propagation matrix A as a primitive, but the construction of the empirical object from the available data is not trivial. We develop a construction that is defensible under a simple microfoundation, connecting individual-level social interactions to state-level opinion dynamics. The argument has two steps: first, we derive the state-level form of A from an individual population-level model. Second, we show that the unknown scale of the SCI matrix is cancelled through row-normalization, so that A is identified directly from the observable aggregate data.

We assume that individual initial opinions are distributed around the state-level average

$$y_{0,i} = x_{0,s(i)} + \epsilon_i,$$

where the shocks ϵ_i are independent across individuals and social ties, mean zero, and have finite variance σ_ϵ^2 . Opinions follow a DeGroot updating process at the individual level: $y_{t+1,i} = \sum_j T_{ij}^{\text{ind}} y_{t,j}$, where $T_{ij}^{\text{ind}} = A_{ij}^{\text{ind}} / d_i^{\text{ind}}$ is the row-stochastic normalization of the individual adjacency matrix.

Social ties form according to a state-level stochastic block model: conditional on state memberships, $A_{ij}^{\text{ind}} \sim \text{Bernoulli}(p_{s(i),s(j)})$ independently across pairs. The cross-state probability matrix $\mathbf{P} = [p_{ss'}]$ is the structural object that summarizes the model. The next result gives a fixed-date large-population justification for the state-level recursion.

¹⁸Parallel or approximate solvers (Laine and Tomlin, 2019) may reduce this gap for large or sparse networks. Each reported instance is solved serially; independent simulation draws may be distributed across processors.

Proposition S2. Fix the number of states. Let $N = \sum_s N_s$, $N_{\min} = \min_s N_s$, and suppose $N_{\min}/N$ is bounded away from zero. Suppose also that there is a constant $\underline{\mu} > 0$ such that every expected individual degree satisfies $\mu_s \geq \underline{\mu}N$. Let

$$x_{t,s}^{(N)} = \frac{1}{N_s} \sum_{i:s(i)=s} y_{t,i}$$

denote the realized state-average opinion. Define

$$\hat{A}_{ss'}^{(N)} = \frac{p_{ss'} N_{s'}}{\mu_s}, \quad \mu_s = \sum_{s''=1}^n p_{ss''} N_{s''}, \quad (\text{S6})$$

and let $\bar{\mathbf{x}}_0 = (x_{0,1}, \dots, x_{0,n})'$. For every fixed date t ,

$$\max_s \left| x_{t,s}^{(N)} - \left((\hat{A}^{(N)})^t \bar{\mathbf{x}}_0 \right)_s \right| \xrightarrow{p} 0 \quad \text{as } N_{\min} \rightarrow \infty.$$

Moreover, $\hat{A}^{(N)}$ is row-stochastic.

Proof of Proposition S2: Let $R^{(N)}$ be the state-averaging matrix, let $P^{(N)} = \mathbb{E}[A^{\text{ind}}]$, and define $\bar{D}^{(N)} = \text{diag}(\mu_{s(1)}, \dots, \mu_{s(N)})$ and $\bar{T}^{(N)} = (\bar{D}^{(N)})^{-1} P^{(N)}$. Thus

$$\bar{T}_{ij}^{(N)} = \frac{p_{ss'}}{\mu_s} \quad \text{when } s(i) = s, \ s(j) = s', \quad R^{(N)} \bar{T}^{(N)} = \hat{A}^{(N)} R^{(N)}.$$

Applying the rectangular Matrix Bernstein inequality to $A^{\text{ind}} - P^{(N)}$ and the scalar Bernstein inequality to the individual degrees gives

$$\|T^{\text{ind}} - \bar{T}^{(N)}\|_2 = O_p \left(\sqrt{\frac{\log N}{N}} \right); \quad (\text{S7})$$

see Tropp (2012, Theorem 1.6) and the related random-graph concentration results in Chung and Radcliffe (2011); Le et al. (2017). The entries of $\bar{T}^{(N)}$ are at most $(\underline{\mu}N)^{-1}$, so $\|\bar{T}^{(N)}\|_2 \leq \underline{\mu}^{-1}$. Equation (S7) then also gives $\|T^{\text{ind}}\|_2 = O_p(1)$. For the fixed date t , the telescoping identity

$$(T^{\text{ind}})^t - (\bar{T}^{(N)})^t = \sum_{\ell=0}^{t-1} (T^{\text{ind}})^\ell (T^{\text{ind}} - \bar{T}^{(N)}) (\bar{T}^{(N)})^{t-1-\ell}.$$

It follows that

$$\|(T^{\text{ind}})^t - (\bar{T}^{(N)})^t\|_2 = O_p\left(\sqrt{\frac{\log N}{N}}\right) = o_p(1).$$

Writing the initial individual-opinion vector as $\mathbf{y}_0 = L^{(N)}\bar{\mathbf{x}}_0 + \boldsymbol{\epsilon}$, where $L^{(N)}$ replicates each state mean across its residents, gives

$$\mathbf{x}_t^{(N)} - (\hat{A}^{(N)})^t \bar{\mathbf{x}}_0 = R^{(N)}[(T^{\text{ind}})^t - (\bar{T}^{(N)})^t] \mathbf{y}_0 + (\hat{A}^{(N)})^t R^{(N)} \boldsymbol{\epsilon}.$$

Now $\|R^{(N)}\|_2 = N_{\min}^{-1/2}$ and $\|\mathbf{y}_0\|_2 = O_p(\sqrt{N})$, so the first term is $o_p(1)$. The s th component of $R^{(N)} \boldsymbol{\epsilon}$ has variance σ_{ϵ}^2/N_s , and hence

$$\|R^{(N)} \boldsymbol{\epsilon}\|_2 = O_p(N_{\min}^{-1/2}) = o_p(1).$$

Since $(\hat{A}^{(N)})^t$ is row-stochastic and n is fixed, its operator norm is bounded, so the second term is also $o_p(1)$. Finally,

$$\sum_{s'} \hat{A}_{ss'}^{(N)} = \frac{\sum_{s'} p_{ss'} N_{s'}}{\mu_s} = 1,$$

which proves row-stochasticity. □

Remark S1. Proposition S2 is an asymptotic, fixed-date result and does not provide a finite-sample error constant. Large state populations make state-average idiosyncratic-opinion noise small, but the theorem alone does not quantify the exact magnitude of the approximation error of the aggregated propagation matrix in a finite sample.

Remark S2. The construction does not require $\hat{A}$ and $\mathbf{x}_0$ to be independent. For example, states may have latent characteristics $\mathbf{v}_s$ that jointly determine average opinions, $x_{0,s} = f_0(\mathbf{v}_s) + \epsilon_s$, and link propensities, $p_{ss'} = W(\mathbf{v}_s, \mathbf{v}_{s'})$. Proposition S2 conditions on these state characteristics, so correlation between initial opinions and network connectivity does not alter the aggregation argument.

Proposition S2 shows that aggregate opinion dynamics can be approximated by the row-normalized state-level matrix $\hat{A}^{(N)}$ constructed from the cross-state probability matrix $\mathbf{P} = [p_{ss'}]$ and state populations. We model the SCI as proportional to the underlying link probabilities. The following result shows that the unknown common scale does not affect

the resulting propagation matrix.

Lemma S1. *Suppose $SCI_{ss'} = \kappa \cdot p_{ss'}$ for an unknown scalar $\kappa > 0$ that is common across state pairs. Then $\hat{A}^{(N)}$ defined in (S6) satisfies*

$$\hat{A}_{ss'}^{(N)} = \frac{SCI_{ss'} N_{s'}}{\sum_{s''=1}^n SCI_{ss''} N_{s''}},$$

so κ cancels identically in the row-normalization. Therefore $\hat{A}^{(N)}$ is point-identified from the observed SCI and state populations, regardless of the scale factor.

Proof of Lemma S1: Since $p_{ss'} = SCI_{ss'}/\kappa$, substituting into (S6) gives

$$\hat{A}_{ss'}^{(N)} = \frac{(SCI_{ss'}/\kappa) N_{s'}}{\sum_{s''} (SCI_{ss''}/\kappa) N_{s''}} = \frac{SCI_{ss'} N_{s'}}{\sum_{s''} SCI_{ss''} N_{s''}}. \quad (\text{S8})$$

The common factor $1/\kappa$ cancels. □

Taken together, Proposition S2 and Lemma S1 provide a fixed-date large-population justification for using the row-normalized, population-weighted SCI matrix as the empirical counterpart of the model primitive A . Following Bailey et al. (2021), we aggregate county-level SCI to states according to

$$SCI_{ab} = \sum_{i \in C(a), j \in C(b)} \frac{\text{Pop}_i}{\text{Pop}_a} \frac{\text{Pop}_j}{\text{Pop}_b} SCI_{ij}.$$

This population-weighted average preserves proportionality to the state-pair link probability with the same scale κ . We then multiply each destination column by its state population and row-normalize as in Lemma S1. The resulting matrix is generally asymmetric because destination populations and row-normalization denominators differ across states, even though the underlying SCI is symmetric. We retain self-links, which are removed from Figure 5 for visualization clarity.

Opinions are county `consensusOppose` estimates from the 2023 Yale Climate Opinion Maps (Howe et al., 2015; Marlon et al., 2022), aggregated to the 50 states and the District of Columbia using county population shares. The network uses 2020 county populations from `co-est2020.csv`; Q^{pop} and the opinion aggregation use the corresponding 2020 population vintage in the replication code. Jurisdictions are ordered alphabetically by postal

abbreviation, including DC. All state pairs are observed, no row has zero mass, self-links are retained in $\hat{A}$, and each row is normalized to one, justified by Proposition S2. As a

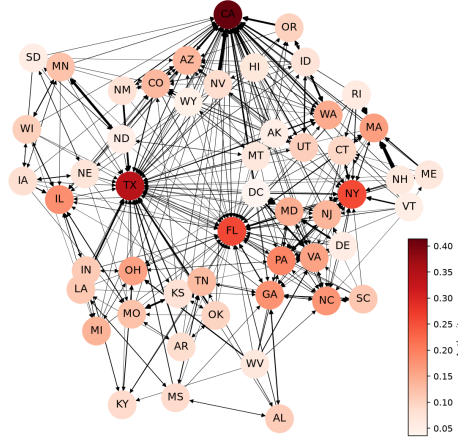


Figure 5: Interstate Facebook friendship network used to construct $\hat{A}$. Nodes are colored by authority, measured by the normalized leading right singular vector of $\hat{A}$; self-links are omitted for visualization.

specification check, we replace the county-aggregated opinions with the published state estimates in the same Yale workbook. The California–Texas exposure gap falls from 0.0032 to 0.0009. Coarse capture remains 99.91%, effective dimension is 1.03, and the leading share is 98.68%. The workbook does not provide state-specific standard errors or a covariance matrix, so we treat this as a specification sensitivity rather than formal statistical inference. The replication script records the source files and regenerates the reported tables and figures.

S5.2 Computation and robustness

We compute the reported population-weighted coarse candidate using the adaptive projected-gradient method in Algorithm 1. The multi-start search uses six informed initializations, $Q\hat{A}\mathbf{x}_0$, $\hat{A}\mathbf{x}_0$, $\mathbf{x}_0$, authority, hub, and the uniform direction, together with random seeds $0, \dots, 5$. All twelve starts reach the same objective to ten decimal places. A separate sphere-constrained optimization changes the exposure direction by less than 1.5×10^{-6} and leaves all reported values unchanged at the displayed precision. The undamped fixed-point

method converges in the opening example and the $Q = I$ application but cycles under Q^{pop} , so it is not used for the primary empirical results. We retain the highest-value candidate and compare it with every reported fixed heuristic. Full targeting uses the unrestricted vector intervention $\mathbf{u}_t$ defined in Section 2. For any payoff matrix Q , the no-intervention loss matrix solves $K^\varnothing = Q + \delta A' K^\varnothing A$.

The primary specification uses $Q = Q^{\text{pop}}$. Authority and hub centrality are the leading right and left singular vectors of $\hat{A}$. One period denotes one application of the state updating matrix, not a calibrated calendar interval. The parameters δ and c are chosen to be illustrative (rather than calibrated), motivating the reported parameter grid. Because $\|\mathbf{b}\|_2 = 1$, $\|\mathbf{u}_t\|_2 = |r_t|$, so c prices squared Euclidean intervention magnitude relative to the normalized opinion-loss scale. Under the baseline sign normalization, all coordinates of the reported exposure vector are positive and the optimal intensity does not reverse sign along the realized path. For completeness, Table 4 reports the wider cost grid and the additional fixed heuristics omitted from the main text.

Table 4: Policy comparison across intervention costs ($\delta = 0.9$, $Q = Q^{\text{pop}}$). Entries report the share of attainable gains captured. Entries displayed as 100.0% are rounded.

	$c = 1$	$c = 10$	$c = 100$	$c = 1000$	$c = 10000$
Opinion ($\mathbf{b} \propto \mathbf{x}_0$)	98.3%	91.1%	72.0%	51.3%	45.2%
Network-updated ($\mathbf{b} \propto \hat{A}\mathbf{x}_0$)	98.5%	91.7%	73.3%	53.0%	46.9%
Payoff-updated ($\mathbf{b} \propto Q^{\text{pop}}\hat{A}\mathbf{x}_0$)	94.3%	97.8%	99.4%	99.4%	99.3%
Authority ($\mathbf{b} \propto \mathbf{v}_1$)	97.2%	99.1%	97.8%	94.2%	92.6%
Static exposure	99.6%	99.2%	99.5%	99.4%	99.3%
Period-by-period myopic retargeting	99.7%	91.5%	52.2%	25.1%	20.2%
Coarse ($\mathbf{b} = \hat{\mathbf{b}}$)	99.9%	99.9%	100.0%	100.0%	100.0%

Table 5 reports the largest state-level differences between the reusable and static exposure at the baseline.

Table 6 reports robustness to discount factors away from the baseline $\delta = 0.9$; Table 4 reports the baseline cost grid in greater detail. The last column uses a separate raw-value normalization, $|v^c - v^f|/|v^f|$, presenting the value gap as a proportion of full targeting value. Through this lens, the coarse rule stays within 0.71% of full targeting under all tested parameter values. The capture columns show that the computed coarse rule dominates every reported fixed heuristic at every cell, while each individual heuristic has a regime where it

Table 5: Reusable and static exposure at the baseline ($\delta = 0.9$, $c = 10$, $Q = Q^{\text{pop}}$). Both directions have unit Euclidean norm. Absolute differences are computed from the unrounded exposure weights.

State	Reusable	Static	Absolute difference
California	0.3280	0.4366	0.1086
Texas	0.3248	0.3983	0.0735
Mississippi	0.0921	0.0604	0.0317
Minnesota	0.1337	0.1025	0.0313
Alabama	0.1331	0.1022	0.0309

falls short. Table 7 reports population-weighted capture under recurrent shocks of size σ

Table 6: Population-weighted capture $(v(\mathbf{b}) - v^{\emptyset})/(v^f - v^{\emptyset})$ across (δ, c) ($Q = Q^{\text{pop}}$). Full targeting is the 100% benchmark; entries displayed as 100.0% are rounded. The last column is the coarse-to-full gap $|v^c - v^f|/|v^f|$.

δ	c	Coarse	Authority	Opinion	Hub	coarse/full gap
0.80	1	99.82%	95.39%	96.14%	97.60%	0.45%
0.80	10	99.87%	98.49%	81.64%	94.64%	0.13%
0.80	100	99.99%	95.80%	56.79%	82.33%	0.00%
0.95	1	99.95%	98.49%	99.20%	99.26%	0.65%
0.95	10	99.95%	99.48%	95.67%	98.80%	0.30%
0.95	100	100.0%	98.88%	84.48%	94.77%	0.01%
0.99	1	99.99%	99.68%	99.85%	99.84%	0.71%
0.99	10	99.99%	99.89%	99.16%	99.77%	0.36%
0.99	100	100.0%	99.78%	96.73%	98.96%	0.01%

at $\delta = 0.9$ and $c = 10$. For each σ , we recompute the coarse direction using the expected discounted opinion-path matrix, which adds the shock covariance to the deterministic path. Both fixed-direction rules lose out to full targeting as σ grows. The table is computed analytically from expected discounted losses under shock covariance $\sigma^2 I$; it does not use Monte Carlo simulation.

S5.3 Effective dimension of the opinion path

Table 8 reports the concentration of the population-weighted realized path $Z(\hat{\mathbf{b}})$ across the parameter grid, underlying the effective-dimension discussion in Section 6.3. At the

Table 7: Population-weighted capture under recurrent shocks of size σ (with $\delta = 0.9$, $c = 10$, $Q = Q^{\text{pop}}$). Full targeting is the 100% benchmark at every σ .

σ	0	0.1	0.2	0.3	0.5	1.0
Coarse ($\hat{\mathbf{b}}$)	99.9%	97.2%	91.3%	85.6%	78.2%	72.4%
Authority ($\mathbf{v}_1$)	99.1%	96.5%	90.8%	85.1%	77.7%	71.6%

baseline, the leading direction carries 98.59% of path weight but accounts for only 1.14% of the realized coarse-to-full gap:

$$\frac{\sigma_1 \mathbf{q}_1' \Phi \mathbf{q}_1}{\text{tr}(\Phi Z)} = 0.0114.$$

The effective path dimension stays near one throughout, so the path remains low-dimensional regardless of (δ, c) . Let $\bar{\phi} = \text{tr}(\Phi)/51$. The penultimate column reports the intervention error on the dominant unit eigenvector $v_{\max}$ of Z relative to this mean, $v'_{\max} \Phi v_{\max} / \bar{\phi}$. The final column reports the realized coarse-to-full loss as a share of attainable gains.

Table 8: Concentration of the population-weighted realized opinion path $Z(\hat{\mathbf{b}})$, network of the 50 states and the District of Columbia ($Q = Q^{\text{pop}}$). “Lead share” is the largest eigenvalue of Z over its trace; $d_{\text{eff}} = (\sum_k \sigma_k)^2 / \sum_k \sigma_k^2$; $v'_{\max} \Phi v_{\max} / \bar{\phi}$ is the intervention error on Z ’s leading direction relative to its mean.

δ	c	Z lead share	d_{eff}	$v'_{\max} \Phi v_{\max} / \bar{\phi}$	Δ/gain
0.80	1	96.72%	1.07	9.0×10^{-4}	0.185%
0.80	10	99.02%	1.02	1.8×10^{-4}	0.127%
0.80	100	99.79%	1.00	6.5×10^{-7}	0.007%
0.90	1	96.14%	1.08	1.3×10^{-3}	0.101%
0.90	10	98.59%	1.03	3.2×10^{-4}	0.087%
0.90	100	99.75%	1.01	2.0×10^{-6}	0.007%
0.95	1	95.86%	1.09	1.5×10^{-3}	0.052%
0.95	10	98.34%	1.03	4.1×10^{-4}	0.050%
0.95	100	99.69%	1.01	4.7×10^{-6}	0.005%
0.99	1	95.63%	1.09	1.7×10^{-3}	0.011%
0.99	10	98.11%	1.04	5.1×10^{-4}	0.011%
0.99	100	99.61%	1.01	1.0×10^{-5}	0.001%

To separate network structure from the placement of observed disagreement, we compare the

Facebook propagation matrix with random row-stochastic networks. The random networks have independent rows drawn from $\text{Dirichlet}(0.02\mathbf{1}_{51})$. The concentration parameter 0.02 is chosen so that the ensemble has persistence comparable to the Facebook network: in a calibration using 200 independently generated networks, the median second-eigenvalue modulus is 0.77, compared with 0.76 for $\hat{A}$.

For the comparison in Table 9, we use 20 fixed seeds, 20260813 through 20260832. For each seed, we generate one random permutation of the empirical disagreement vector $\mathbf{d}$ and one random network from the calibrated ensemble. This gives four environments: $(\hat{A}, \mathbf{d})$, $(\hat{A}, \mathbf{d}^{(s)})$, $(A^{(s)}, \mathbf{d})$, and $(A^{(s)}, \mathbf{d}^{(s)})$. The same permutation is used in the second and fourth environments and the same random network in the third and fourth. We hold $Q = Q^{\text{pop}}$, $\delta = 0.9$, and $c = 10$ fixed and reoptimize the coarse rule in every environment.

Table 9: Network and opinion sources of centered path concentration. Randomized rows report medians; brackets give the 10th and 90th percentiles.

Environment	$d_{\text{eff}}(Z)$	Lead share	Coarse capture
Facebook $\hat{A}$, observed $\mathbf{d}$	1.08	96.24%	99.61%
Facebook $\hat{A}$, permuted $\mathbf{d}$	1.04 [1.03, 1.07]	98.02% [96.83, 98.66]	98.97% [98.28, 99.47]
Random A , observed $\mathbf{d}$	2.16 [1.83, 2.43]	63.77% [58.12, 71.08]	78.77% [71.30, 87.46]
Random A , permuted $\mathbf{d}$	2.13 [1.82, 2.56]	63.90% [56.22, 71.80]	83.75% [70.44, 90.29]

The Facebook network produces a nearly one-dimensional path even when the observed disagreement vector is permuted. Random networks produce paths that are substantially less concentrated under either the observed or permuted disagreement. These diagnostics indicate that the concentration is primarily a feature of the Facebook propagation matrix rather than a special alignment between that matrix and the observed ordering of $\mathbf{d}$. The observed pairing does, however, raise coarse capture relative to the median permutation on the Facebook network.